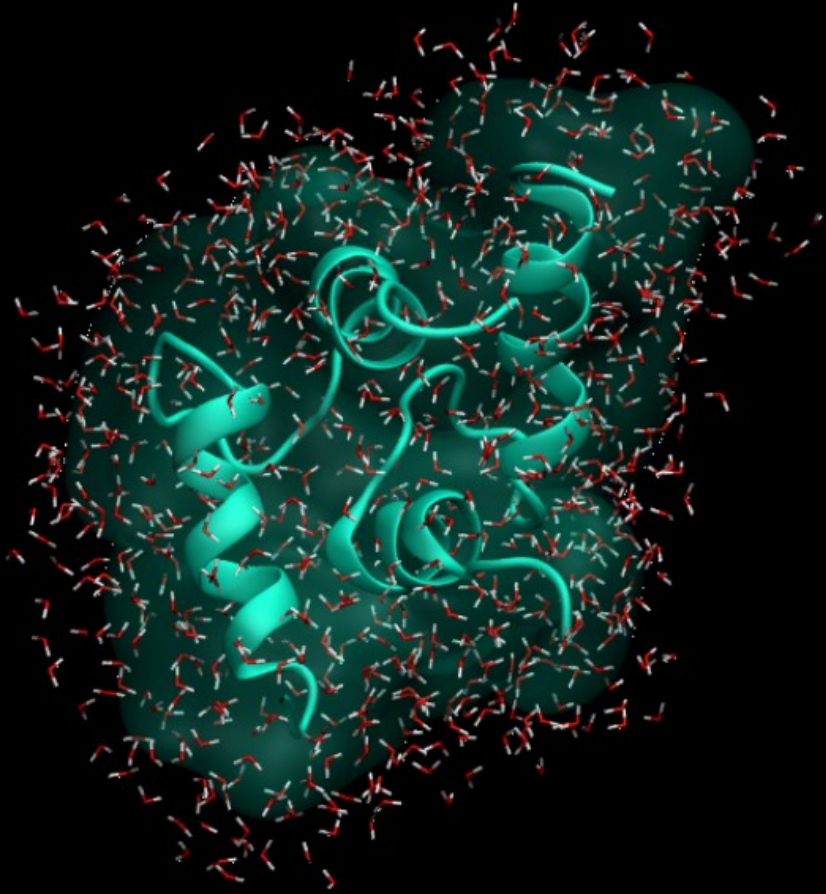




# Molecular dynamics

## Practical session

Simon Gravelle  
LIPhy (Grenoble France), CNRS

Download LAMMPS-GUI :

<https://github.com/simongravelle/PHELMA>

📖 README



## LAMMPS files | DIADEM

---

### Download LAMMPS-GUI

---

Alternatively, LAMMPS-GUI can also be downloaded from these links:

- [LAMMPS-GUI \(.exe\)](#) for Windows
- [LAMMPS-GUI \(.tar.gz\)](#) for Linux

Alternatively, LAMMPS-GUI can be downloaded from <https://github.com/akohlmeier/lammps-gui/releases/tag/v3.0.7>.

### Problem with opening LAMMPS-GUI ?

---

See [this page](#)

File Edit Run View Tutorials About

```
1 # When using LAMMPS-GUI in your project, please cite: https://doi.org/10.33011/livecoms.6.1.3037  
2 |
```



# <https://github.com/simongravelle/PHELMA>

## First input

```
# Simple NVE argon simulation
units          lj # Use Lennard-Jones reduced units
dimension      3 # Perform the simulation in 3 spatial dimensions
atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
boundary       p p p # Use periodic boundary conditions

region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
create_box     1 simbox # Create a simulation box containing one atom type
create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box

mass           1      1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
pair_coeff     1      1      1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0)

fix            mynve      all      nve # Integrate the equations of motion
timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units

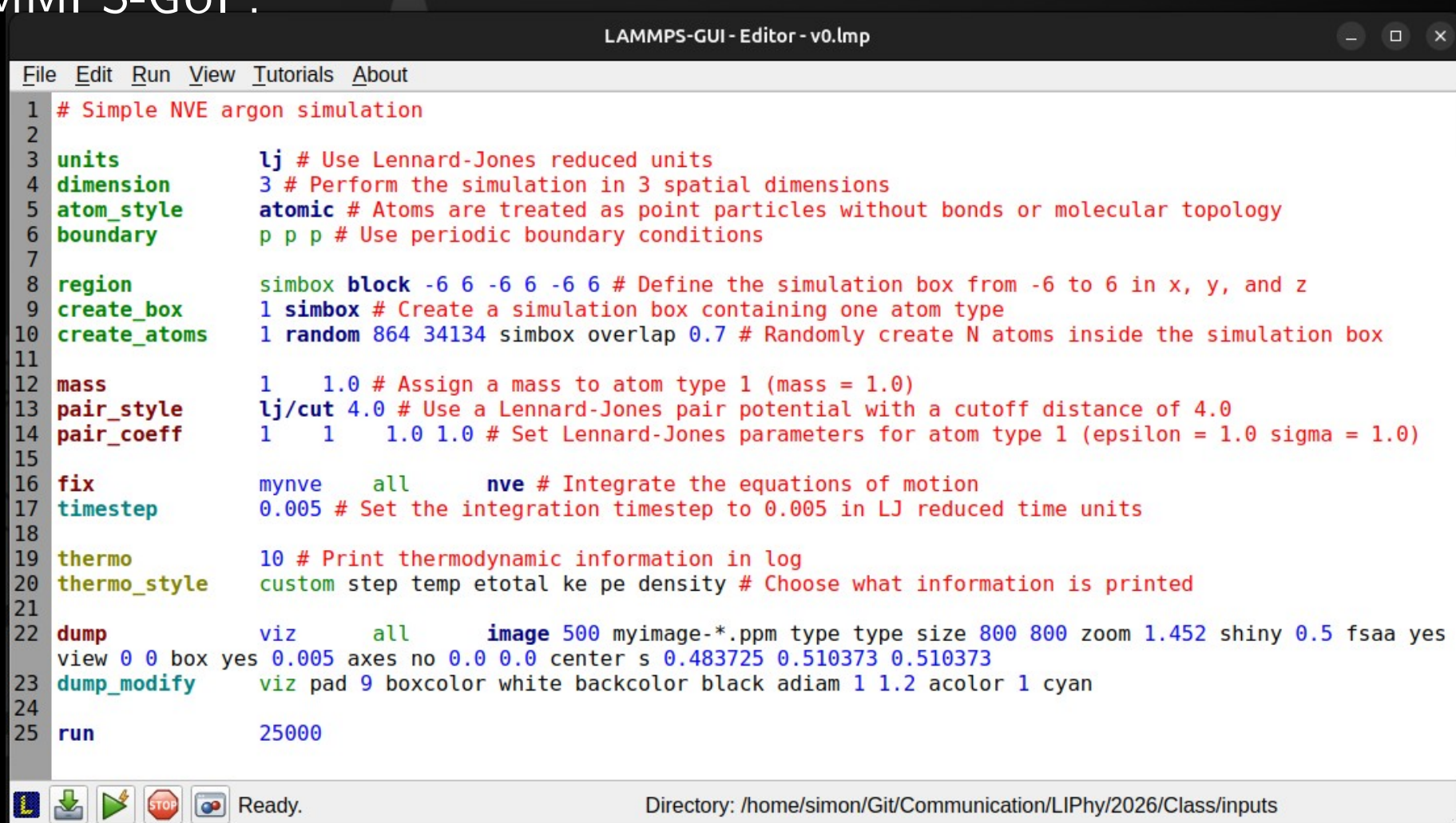
thermo         10 # Print thermodynamic information in log
thermo_style   custom step temp etotal ke pe density # Choose what information is printed

dump           viz        all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shift 0
dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan

run            25000
```

Copy input from <https://github.com/simongravelle/PHELMA> into the LAMMPS-GUI :

Copy input from <https://github.com/simongravelle/PHELMA> into the LAMMPS-GUI :

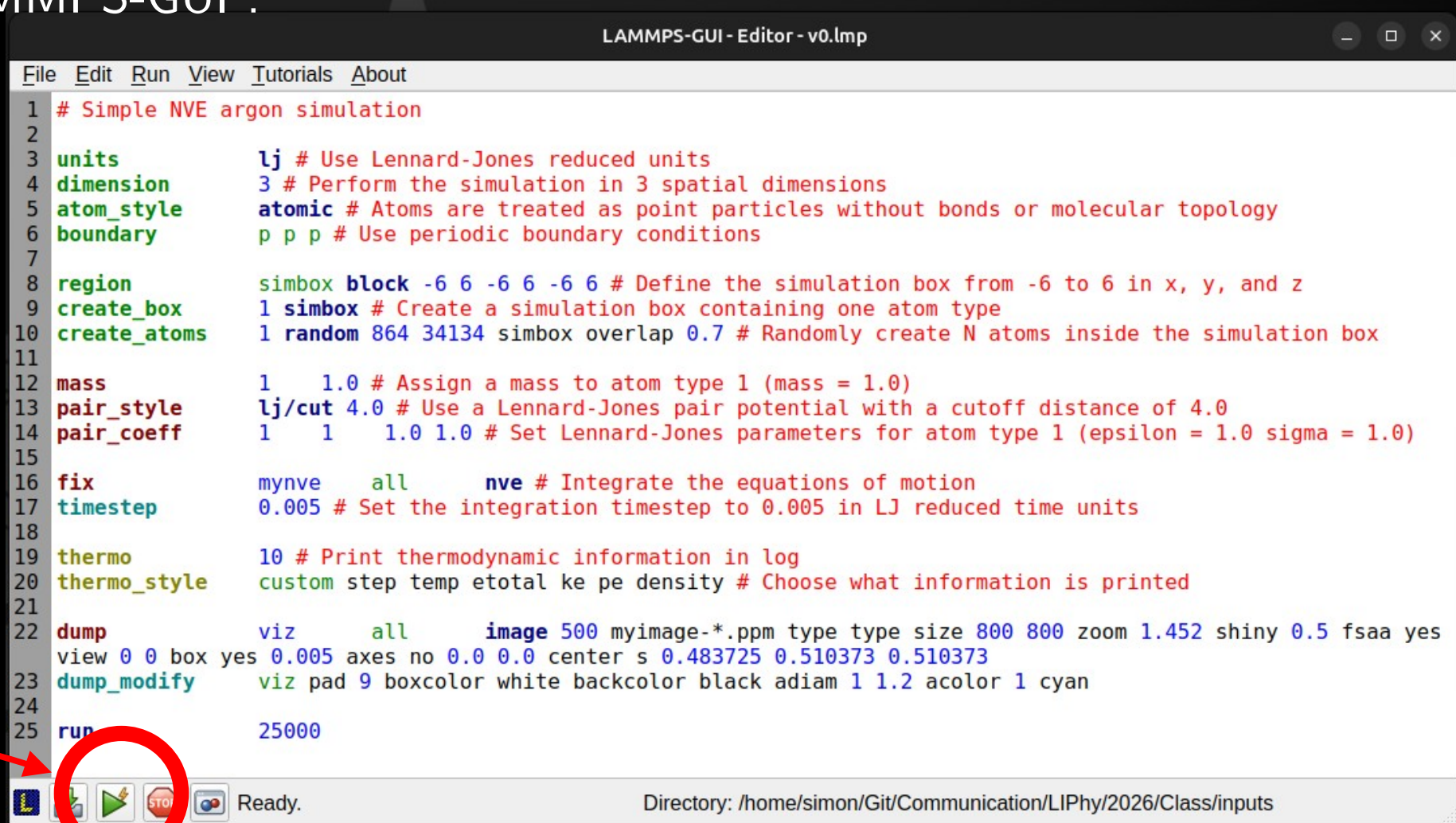


The screenshot shows the LAMMPS-GUI Editor window titled "LAMMPS-GUI - Editor - v0.lmp". The window has a menu bar with "File", "Edit", "Run", "View", "Tutorials", and "About". The main text area contains a script for a simple NVE argon simulation. The script is as follows:

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run           25000
```

The status bar at the bottom shows icons for file operations and the text "Ready." The directory path is displayed as "Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs".

Copy input from <https://github.com/simongravelle/PHELMA> into the LAMMPS-GUI :



The screenshot shows the LAMMPS-GUI Editor window with the following input script:

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify   viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run          25000
```

The bottom toolbar contains several icons. A red arrow points to the 'Launch' button, which is a green play button icon. The status bar at the bottom indicates 'Ready.' and the current directory is '/home/simon/Git/Communication/LIPhy/2026/Class/inputs'.

LAMMPS-GUI - Output - v0.lmp - Run 2

|        |         |         |         |     |       |
|--------|---------|---------|---------|-----|-------|
| Pair   | 2.0817  | 2.0817  | 2.0817  | 0.0 | 16.56 |
| Neigh  | 3.5572  | 3.5572  | 3.5572  | 0.0 | 28.30 |
| Comm   | 0.48878 | 0.48878 | 0.48878 | 0.0 | 3.89  |
| Output | 6.2232  | 6.2232  | 6.2232  | 0.0 | 49.51 |
| Modify | 0.18262 | 0.18262 | 0.18262 | 0.0 | 1.45  |
| Other  |         | 0.03689 |         |     | 0.29  |

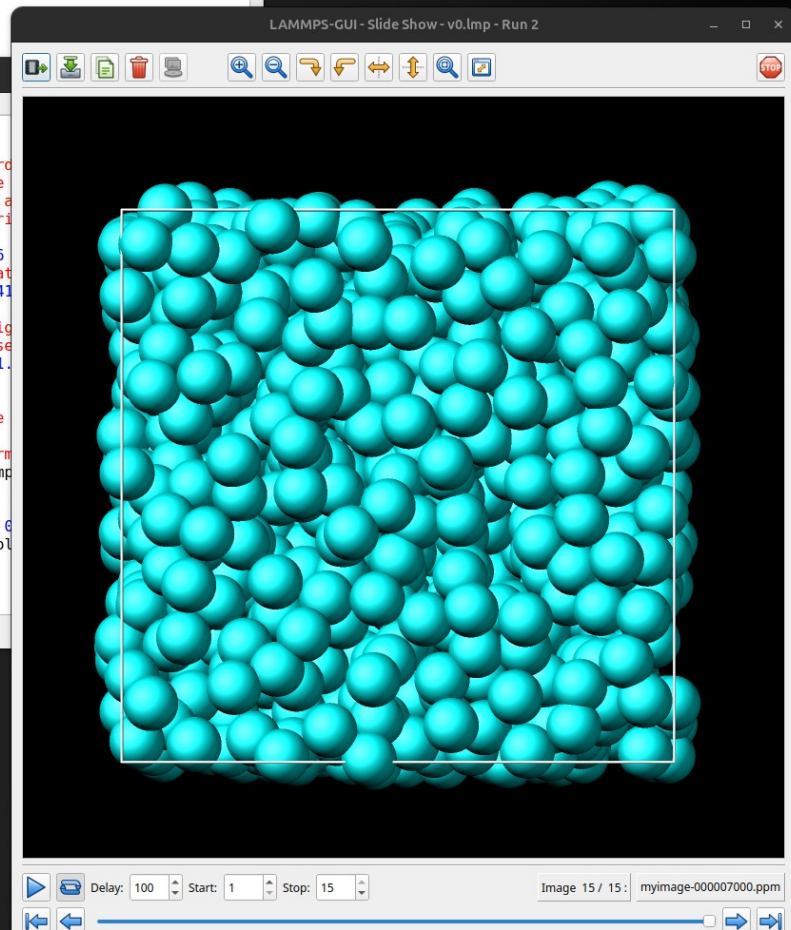
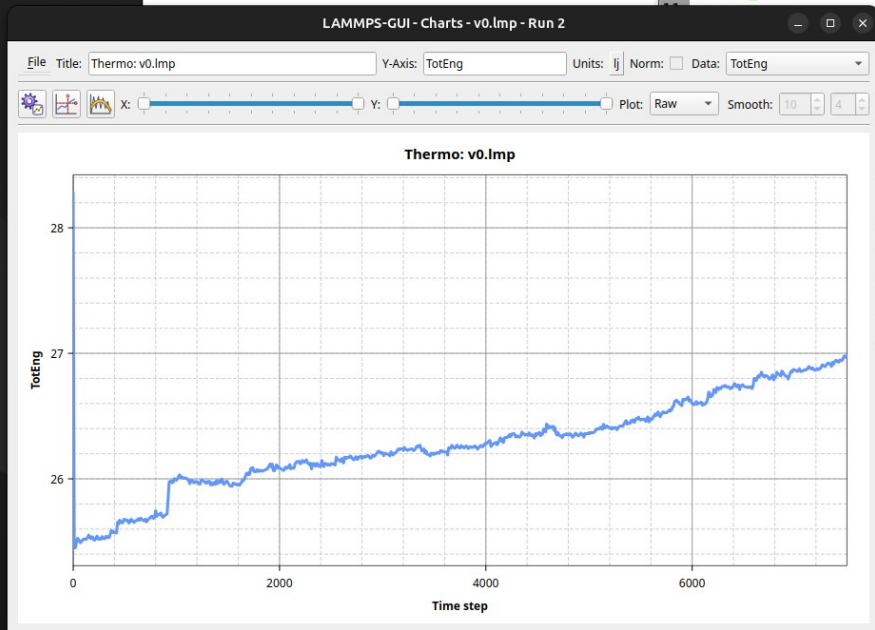
Nlocal: 864 ave 864 max 864 min  
Histogram: 1 0 0 0 0 0 0 0 0  
Nghost: 3510 ave 3510 max 3510 min  
Histogram: 1 0 0 0 0 0 0 0 0  
Neighs: 71640 ave 71640 max 71640 min  
Histogram: 1 0 0 0 0 0 0 0 0

Total # of neighbors = 71640  
Ave neighs/atom = 82.916667  
Neighbor list builds = 3724  
Dangerous builds = 0

File Edit Run View Tutorials About

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard
4 dimension      3 # Perform the
5 atom_style     atomic # Atoms a
6 boundary       p p p # Use peri
7
8 region         simbox block -6
9 create_box     1 simbox # Creat
10 create_atoms   1 random 864 341
```

```
1.0 # Assign
2.0 # Use
3.0 # Use
4.0 # Use
5.0 # Use
6.0 # Use
7.0 # Use
8.0 # Use
9.0 # Use
10.0 # Use
11.0 # Use
12.0 # Use
13.0 # Use
14.0 # Use
15.0 # Use
16.0 # Use
17.0 # Use
18.0 # Use
19.0 # Use
20.0 # Use
21.0 # Use
22.0 # Use
23.0 # Use
24.0 # Use
25.0 # Use
26.0 # Use
27.0 # Use
28.0 # Use
29.0 # Use
30.0 # Use
31.0 # Use
32.0 # Use
33.0 # Use
34.0 # Use
35.0 # Use
36.0 # Use
37.0 # Use
38.0 # Use
39.0 # Use
40.0 # Use
41.0 # Use
42.0 # Use
43.0 # Use
44.0 # Use
45.0 # Use
46.0 # Use
47.0 # Use
48.0 # Use
49.0 # Use
50.0 # Use
51.0 # Use
52.0 # Use
53.0 # Use
54.0 # Use
55.0 # Use
56.0 # Use
57.0 # Use
58.0 # Use
59.0 # Use
60.0 # Use
61.0 # Use
62.0 # Use
63.0 # Use
64.0 # Use
65.0 # Use
66.0 # Use
67.0 # Use
68.0 # Use
69.0 # Use
70.0 # Use
71.0 # Use
72.0 # Use
73.0 # Use
74.0 # Use
75.0 # Use
76.0 # Use
77.0 # Use
78.0 # Use
79.0 # Use
80.0 # Use
81.0 # Use
82.0 # Use
83.0 # Use
84.0 # Use
85.0 # Use
86.0 # Use
87.0 # Use
88.0 # Use
89.0 # Use
90.0 # Use
91.0 # Use
92.0 # Use
93.0 # Use
94.0 # Use
95.0 # Use
96.0 # Use
97.0 # Use
98.0 # Use
99.0 # Use
100.0 # Use
```



File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify   viz pad 9 boxcolor white backcolor black adiam 1 1.2 acolor 1 cyan
25
26 run           25000

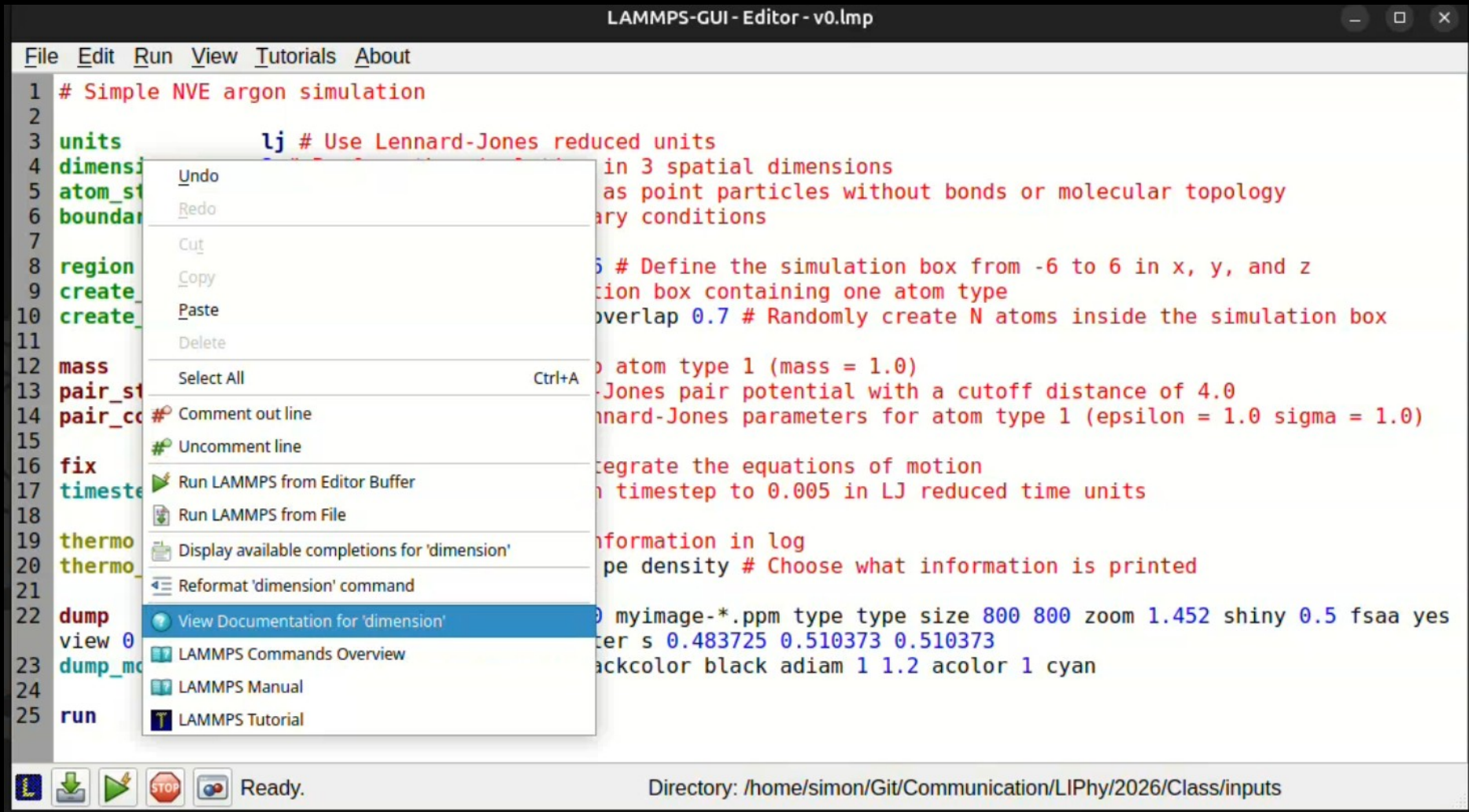
```

Read from top to bottom

File Edit Run View Tutorials About

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass           1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo         10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump           viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white backcolor black adiam 1 1.2 acolor 1 cyan
25
26 run           25000
```

Each line is a command. Everything after # is ignored.



The screenshot shows the LAMMPS-GUI Editor interface. The title bar reads "LAMMPS-GUI - Editor - v0.lmp". The menu bar includes File, Edit, Run, View, Tutorials, and About. The main text area contains a script for a simple NVE argon simulation. A context menu is open over the "dimension" command on line 4, showing options like Undo, Redo, Cut, Copy, Paste, Delete, Select All, and several "Run LAMMPS" and "View Documentation" options. The script text is as follows:

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      in 3 spatial dimensions
5 atom_style     as point particles without bonds or molecular topology
6 boundary       p p p periodic boundary conditions
7
8 region         # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 # Create a simulation box containing one atom type
10 create_atoms   1 # Randomly create N atoms inside the simulation box
11
12 mass           1 # Atom type 1 (mass = 1.0)
13 pair_style     lj # Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff      1 1 1.0 1.0 # Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix            1 # Integrate the equations of motion
17 timestep       0.005 # timestep to 0.005 in LJ reduced time units
18
19 thermo         1 # Display information in log
20 thermo_style   scalar # Choose what information is printed
21
22 dump            1 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view           0 # Center s 0.483725 0.510373 0.510373
24 dump_modify     1 # Blackcolor black adiam 1 1.2 acolor 1 cyan
25
26 run
```

The status bar at the bottom shows icons for file operations and the text "Ready." The directory path is displayed as "/home/simon/Git/Communication/LIPhy/2026/Class/inputs".

File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run           25000

```



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

File Edit Run View Tutorials About

## Generic definitions

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify   viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run          25000

```



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

## Generic definitions

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions

```

$$r^* = \frac{r}{\sigma}$$

$$t^* = \frac{t}{\tau},$$

$$E^* = \frac{E}{\epsilon}$$

$$P^* = \frac{P\sigma^3}{\epsilon}$$

$$\tau = \sigma \sqrt{\frac{m}{\epsilon}}$$

$$U(r) = 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right]$$

File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass           1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo         10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump           viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run           25000

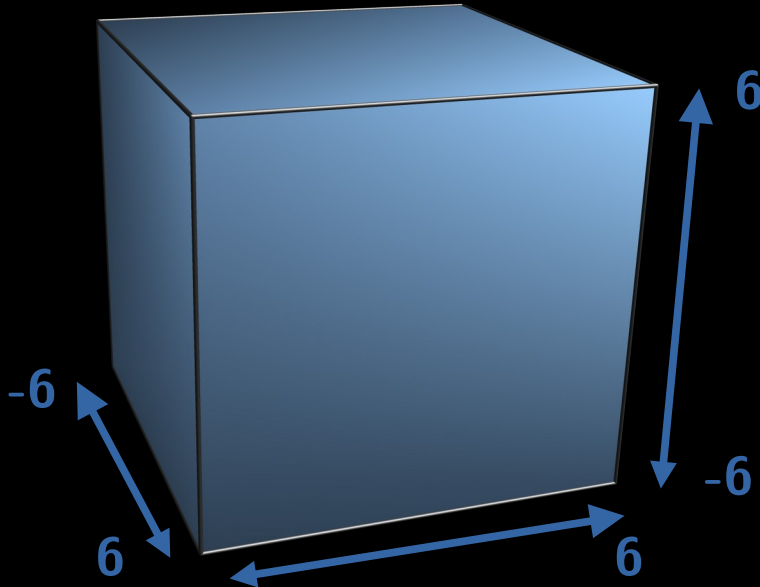
```

## Creation of the system

File Edit Run View Tutorials About

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
```

Creation of the system



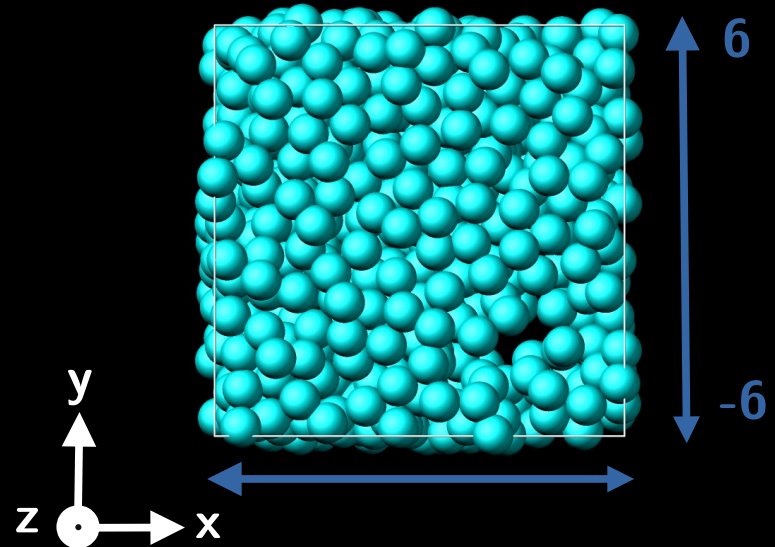
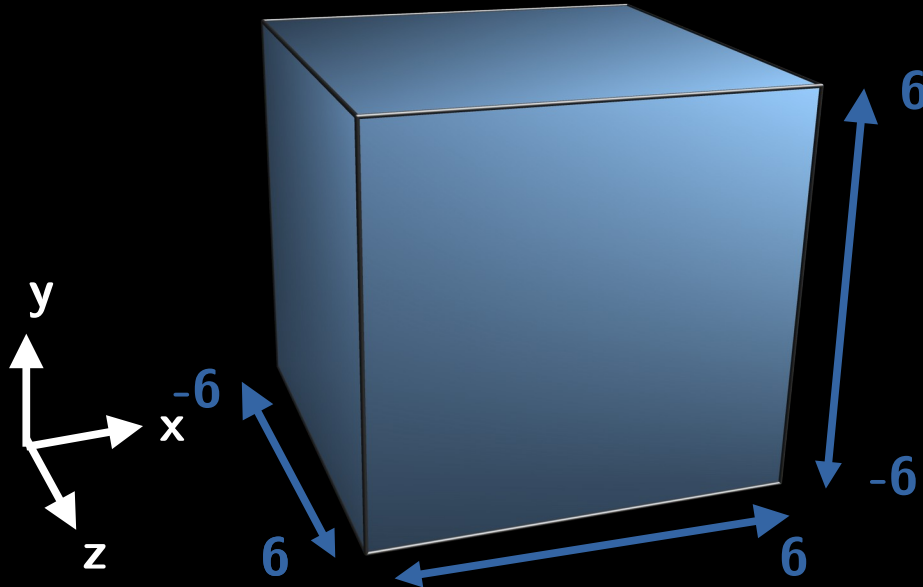
File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11

```

## Creation of the system



File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside
11
12 mass           1      1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1      1      1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve      all      nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump           viz        all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25
26 run           25000

```

Set parameters



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

File Edit Run View Tutorials About

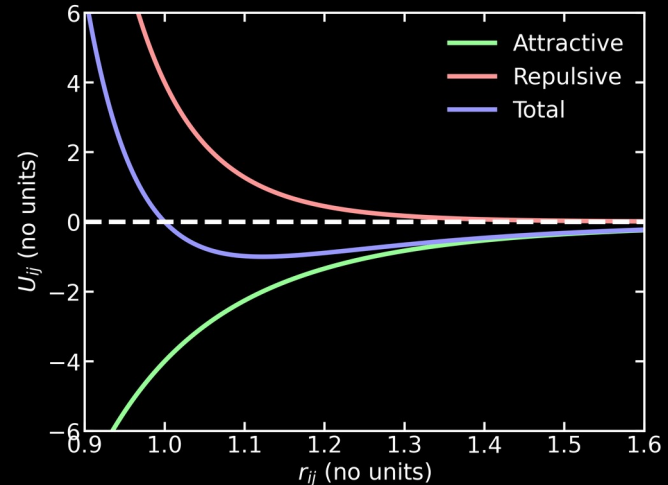
```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15

```

Set parameters

$$U_{ij}(r_{ij}) = 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$



File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff di
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run           25000

```

Equations of motion



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff di
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units

```

Equations of motion

$$\sum_{j \neq i} \mathbf{F}_{ij} = m_i \ddot{\mathbf{r}}_i$$

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

$$\dot{\mathbf{r}}(t + \Delta t) = \dot{\mathbf{r}}(t) + \frac{\ddot{\mathbf{r}}(t) + \ddot{\mathbf{r}}(t + \Delta t)}{2}\Delta t$$

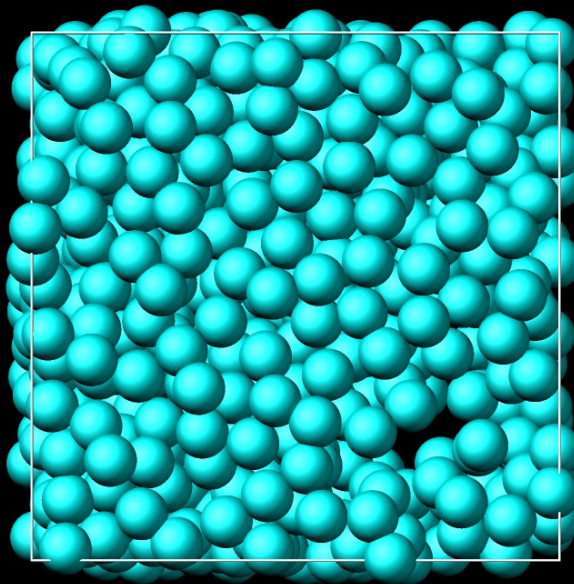
File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass           1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo         10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump           viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25
26 run           25000

```

Visualisation



## Visualisation

```

18
19 thermo          10 # Print thermodynamic information in log
20 thermo_style     custom step temp etotal ke pe density # Choose what information is printed
21
22 dump            viz      all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
23 dump_modify     viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
24
25 run            25000

```



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

File Edit Run View Tutorials About

```

1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms  1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass          1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix           mynve all nve # Integrate the equations of motion
17 timestep      0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo        10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump          viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify   viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run          25000

```

Number of steps



Ready.

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

LAMMPS-GUI - Editor - v0.lmp

File Edit Run View Tutorials About

```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary        p p p # Use periodic boundary conditions
7
8 region          simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box      1 simbox # Create a simulation box containing one atom type
10 create_atoms    1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass            1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff       1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix             mynve all nve # Integrate the equations of motion
17 timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo          10 # Print thermodynamic information in log
20 thermo_style     custom step temp etotal ke pe density # Choose what information is printed
21
22 dump            viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify     viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run            25000
```

Ready. Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

```
LAMMPS-GUI - Editor - v0.lmp
File Edit Run View Tutorials About
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary        p p p # Use periodic boundary conditions
7
8 region          simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box      1 simbox # Create a simulation box containing one atom type
10 create_atoms    1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass            1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff       1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix             mynve all nve # Integrate the equations of motion
17 timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo          10 # Print thermodynamic information in log
20 thermo_style     custom step temp etotal ke pe density # Choose what information is printed
21
22 dump            viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify     viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run            25000
```

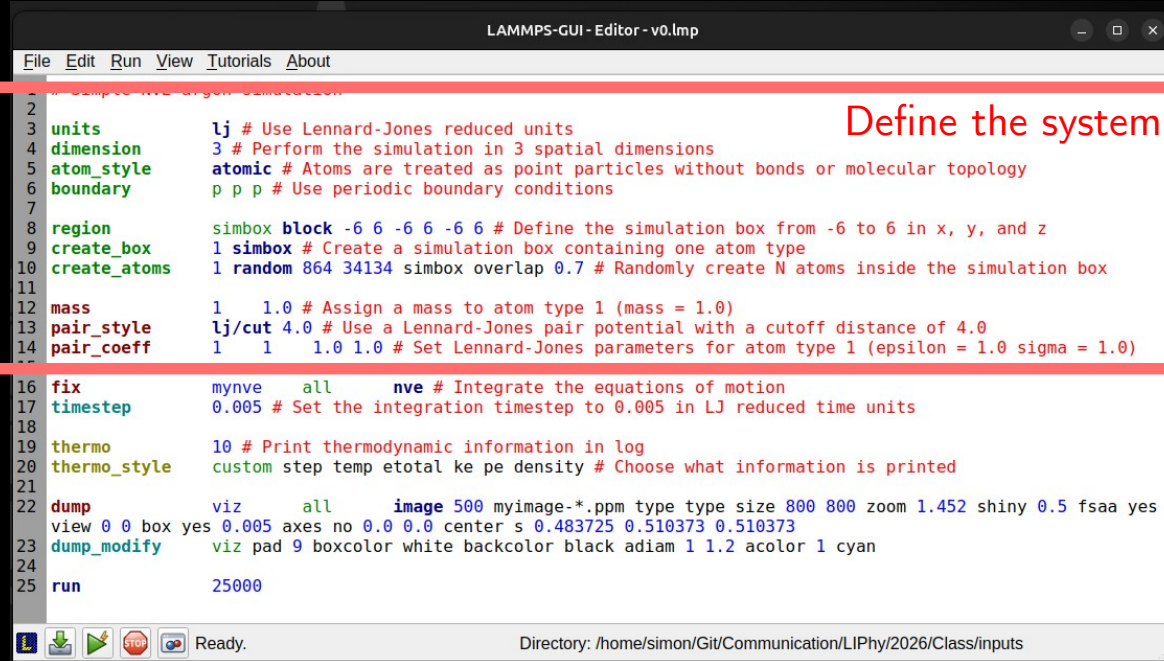
Ready. Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

1) Define the system

2) Measure forces

3) Solve Newton's eq

4) Move atoms



```
File Edit Run View Tutorials About
1 # Sample input script for LAMMPS
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary        p p p # Use periodic boundary conditions
7
8 region          simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box      1 simbox # Create a simulation box containing one atom type
10 create_atoms    1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass            1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff       1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix             mynve all nve # Integrate the equations of motion
17 timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo          10 # Print thermodynamic information in log
20 thermo_style     custom step temp etotal ke pe density # Choose what information is printed
21
22 dump            viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify     viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25
26 run             25000
```

Define the system

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

1) Define the system

2) Measure forces

3) Solve Newton's eq

4) Move atoms

LAMMPS-GUI - Editor - v0.lmp

```
File Edit Run View Tutorials About
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary       p p p # Use periodic boundary conditions
7
8 region         simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box     1 simbox # Create a simulation box containing one atom type
10 create_atoms   1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass           1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff      1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0, sigma = 1.0)
15
16 fix            mynve all nve # Integrate the equations of motion
17 timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo         10 # Print thermodynamic information in log
20 thermo_style   custom step temp etotal ke pe density # Choose what information is printed
21
22 dump           viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify    viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cyan
25 run           25000
```

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

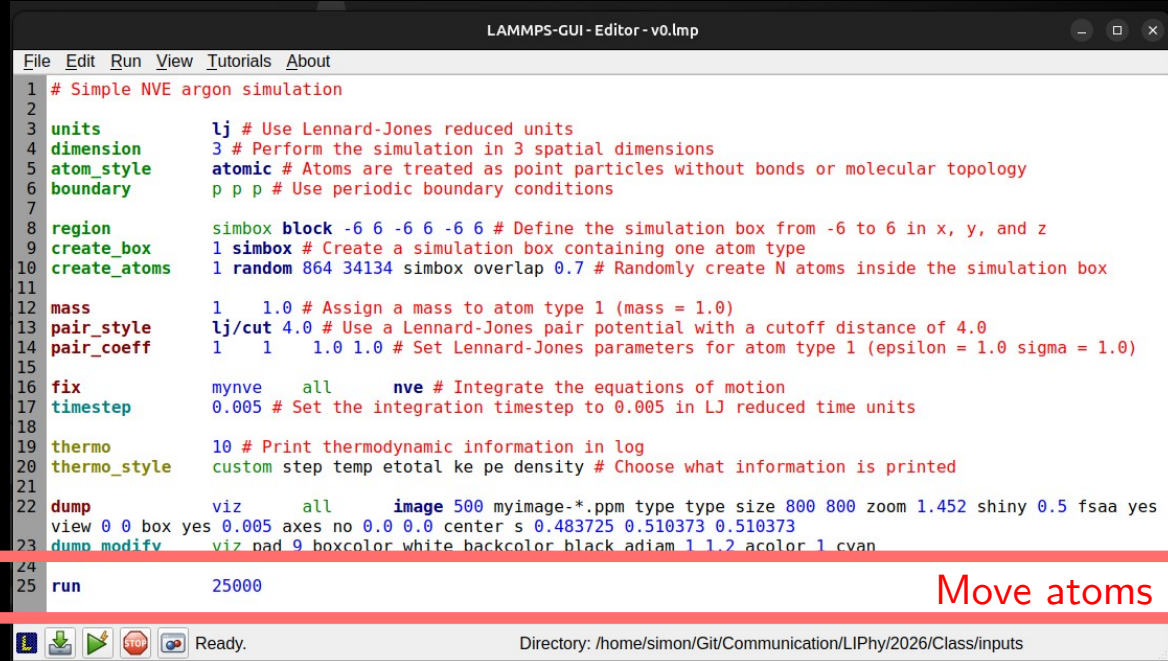
Measure forces  
+ Solve Newton

1) Define the system

2) Measure forces

3) Solve Newton's eq

4) Move atoms



```
1 # Simple NVE argon simulation
2
3 units          lj # Use Lennard-Jones reduced units
4 dimension      3 # Perform the simulation in 3 spatial dimensions
5 atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
6 boundary        p p p # Use periodic boundary conditions
7
8 region          simbox block -6 6 -6 6 -6 6 # Define the simulation box from -6 to 6 in x, y, and z
9 create_box      1 simbox # Create a simulation box containing one atom type
10 create_atoms    1 random 864 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
11
12 mass            1 1.0 # Assign a mass to atom type 1 (mass = 1.0)
13 pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
14 pair_coeff       1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
15
16 fix             mynve all nve # Integrate the equations of motion
17 timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
18
19 thermo          10 # Print thermodynamic information in log
20 thermo_style     custom step temp etotal ke pe density # Choose what information is printed
21
22 dump            viz all image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
23 view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
24 dump_modify     viz pad 9 boxcolor white bgcolor black adiam 1 1.2 acolor 1 cvan
25 run            25000
```

Move atoms

Directory: /home/simon/Git/Communication/LIPhy/2026/Class/inputs

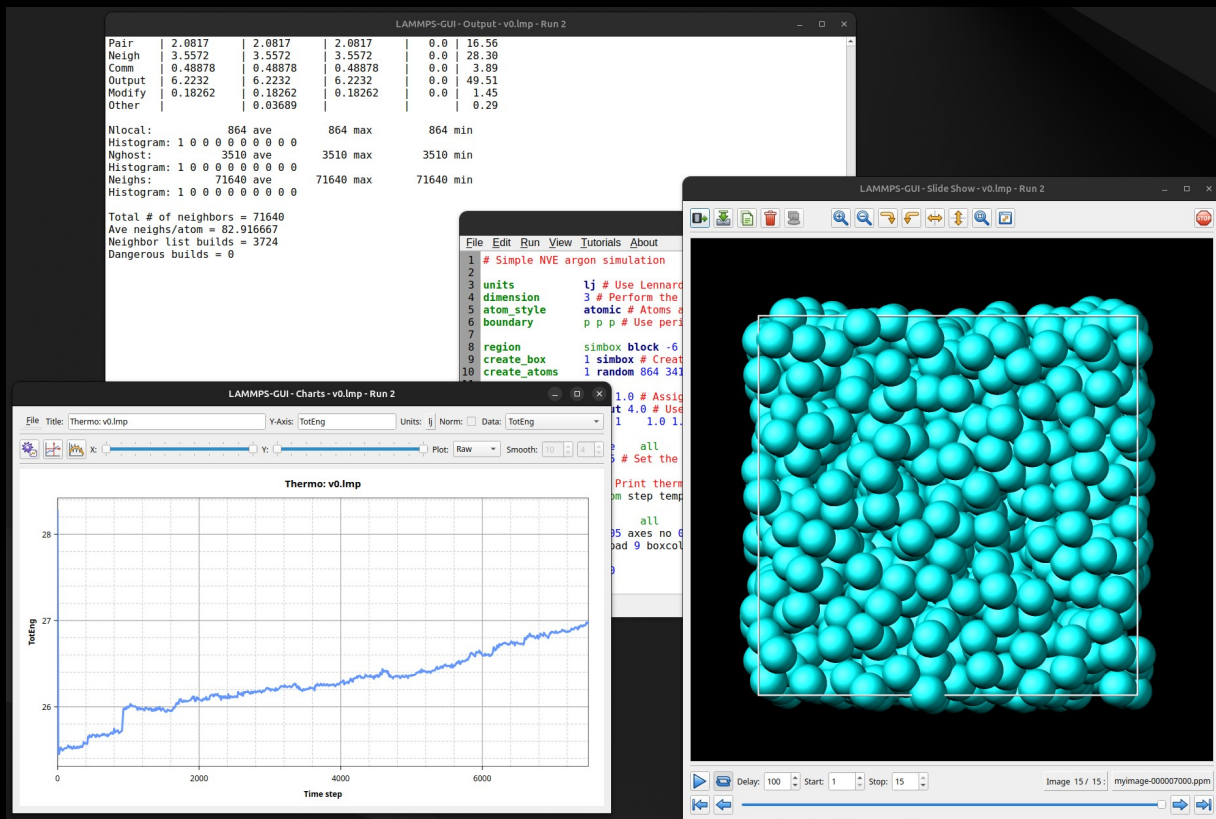
1) Define the system

2) Measure forces

3) Solve Newton's eq

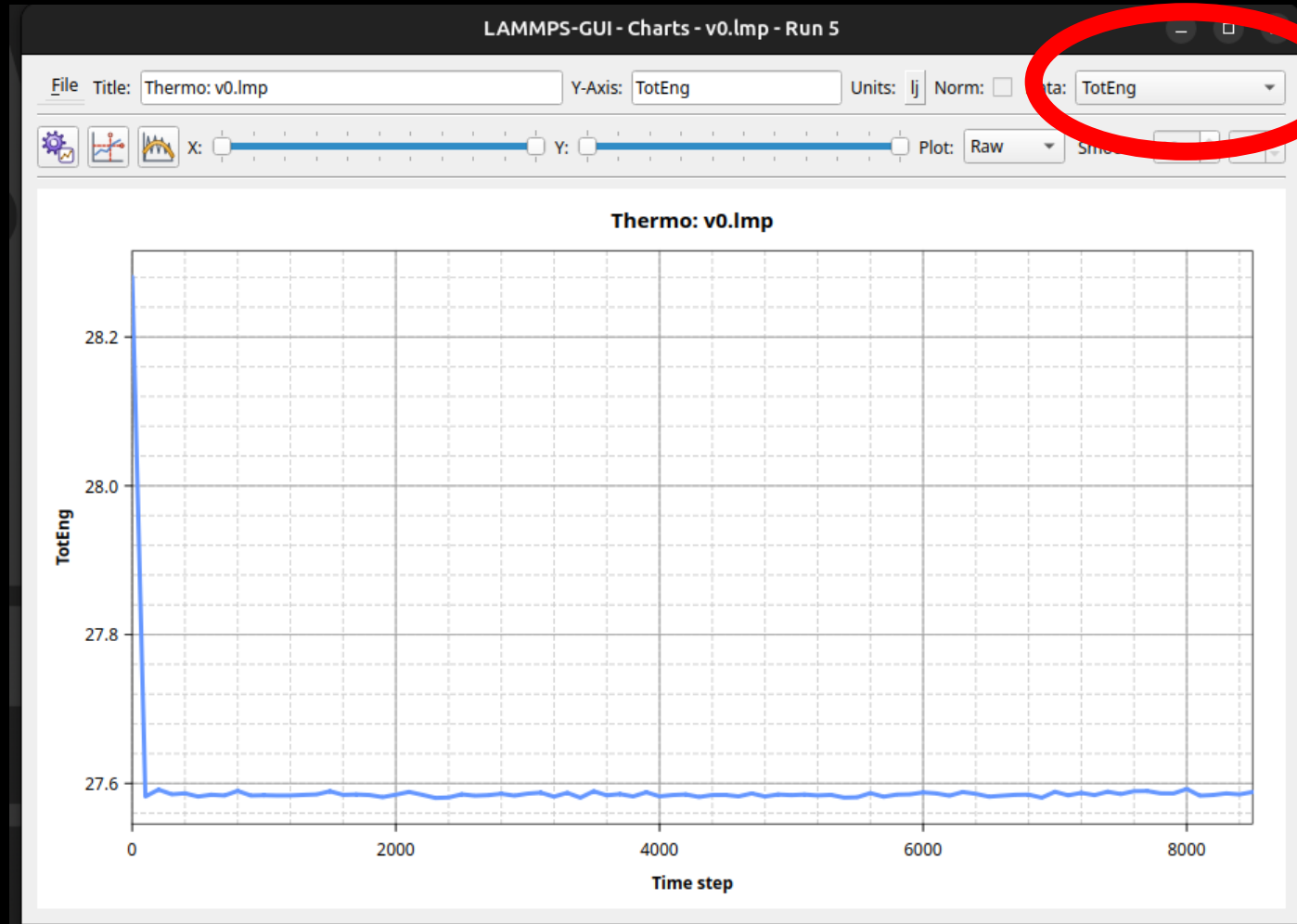
4) Move atoms

# The windows



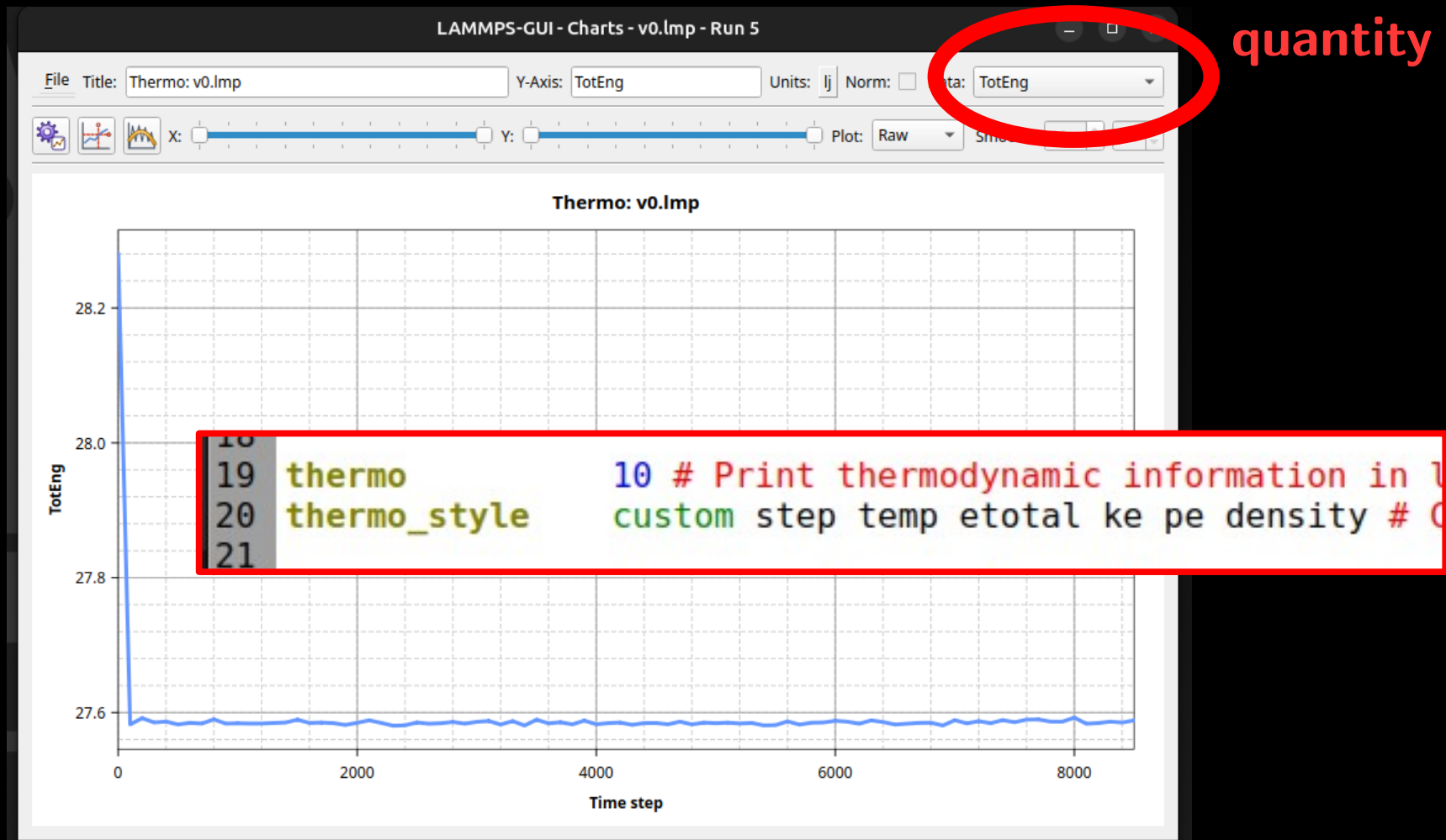
# The charts window

Select different quantity



# The charts window

Select different quantity



# The output window

LAMMPS-GUI - Output - v0.lmp - Run 8

```
using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created orthogonal box = (-6 -6 -6) to (6 6 6)
  1 by 1 by 1 MPI processor grid
Created 864 atoms
  using lattice units in orthogonal box = (-6 -6 -6) to (6 6 6)
  create_atoms CPU = 0.031 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 6 6 6
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
    attributes: half, newton on, omp
    pair build: half/bin/atomonly/newton/omp
    stencil: half/bin/3d
    bin: standard
Setting up Verlet run ...
Unit style      : lj
Current step    : 0
Time step       : 0.0025
Per MPI rank memory allocation (min/avg/max) = 5.852 | 5.852 | 5.852 Mbytes
```

| Step | Temp      | TotEng    | KinEng    | PotEng    | Density |
|------|-----------|-----------|-----------|-----------|---------|
| 0    | 0         | 28.280873 | 0         | 28.280873 | 0.5     |
| 100  | 18.161857 | 27.582391 | 27.2      | 0.382391  | 0.5     |
| 200  | 18.173778 | 27.591709 | 27.2      | 0.391709  | 0.5     |
| 300  | 18.270644 | 27.585356 | 27.374246 | 0.2111091 | 0.5     |

0 Warnings / Errors - 304 Lines 

# The output window

```
LAMMPS-GUI - Output - v0.lmp - Run 8

using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created orthogonal box = (-6 -6 -6) to (6 6 6)
  1 by 1 by 1 MPI processor grid
Created 864 atoms
  using lattice units in orthogonal box = (-6 -6 -6) to (6 6 6)
  create_atoms CPU = 0.031 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 6 6 6
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
    attributes: half, newton on, omp
    pair build: half/bin/atomonly/newton/omp
    stencil: half/bin/3d
    bin: standard
Setting up Verlet run ...
Unit style      : lj
Current step    : 0
Time step       : 0.0025
Per MPI rank memory allocation (min/avg/max) = 5.852 | 5.852 | 5.852 Mbytes
Step    Temp      TotEng      KinEng      PotEng      Density
   0     0      28.280873      0      28.280873      0.5
  100    18.161857  27.582391  27.21111000  0.37128100  0.5
  200    18.173778  27.591709  27.21111000  0.37859700  0.5
  300    18.270644  27.585356  27.51111000  0.07424600  0.5
```

0 Warnings / Errors - 304 Lines 

See errors and warnings

# The output window

LAMMPS-GUI - Output - v0.lmp - Run 8

```
using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created ortho box = (-6 -6 -6) to (6 6 6)
1 by 1 by 1 MPI processor grid
Created 864 atoms
  using lattice units in ortho box = (-6 -6 -6) to (6 6 6)
  create_atoms CPU = 0.031 seconds
Generated mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 6 6 6
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
    attributes: half, newton on, omp
    pair build: half/bin/atomonly/newton/omp
    stencil: half/bin/3d
    bin: standard
Setting up Verlet run ...
Unit style      : lj
Current step    : 0
Time step       : 0.0025
Per MPI rank memory allocation (min/avg/max) = 5.852 | 5.852 | 5.852 Mbytes
```

| Step | Temp      | TotEng    | KinEng    | PotEng     | Density |
|------|-----------|-----------|-----------|------------|---------|
| 0    | 0         | 28.280873 | 0         | 28.280873  | 0.5     |
| 100  | 18.161857 | 27.582391 | 27.2      |            | 0.5     |
| 200  | 18.173778 | 27.591709 | 27.2      |            | 0.5     |
| 300  | 18.270644 | 27.585356 | 27.374246 | 0.21111000 | 0.5     |

0 Warnings / Errors - 304 Lines 

# The output window

LAMMPS-GUI - Output - v0.lmp - Run 8

```
using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created orthogonal box = (-6 -6 -6) to (6 6 6)
  1 by 1 by 1 MPI processor grid
Created 864 atoms
  using lattice units in orthogonal box = (-6 -6 -6) to (6 6 6)
  create_atoms CPU = 0.031 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 6 6 6
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
    attributes: half, newton on, omp
    pair build: half/bin/atomonly/newton/omp
    stencil: half/bin/3d
    bin: standard
Setting up Verlet run ...
Unit style      : lj
Current step    : 0
Time step       : 0.0025
Per MPI rank memory allocation (min/avg/max) = 5.852 | 5.852 | 5.852 Mbytes
```

| Step | Temp      | TotEng    | KinEng    | PotEng     | Density |
|------|-----------|-----------|-----------|------------|---------|
| 0    | 0         | 28.280873 | 0         | 28.280873  | 0.5     |
| 100  | 18.161857 | 27.582391 | 27.2      |            | 0.5     |
| 200  | 18.173778 | 27.591709 | 27.2      |            | 0.5     |
| 300  | 18.270644 | 27.585356 | 27.374246 | 0.21111000 | 0.5     |

0 Warnings / Errors - 304 Lines 

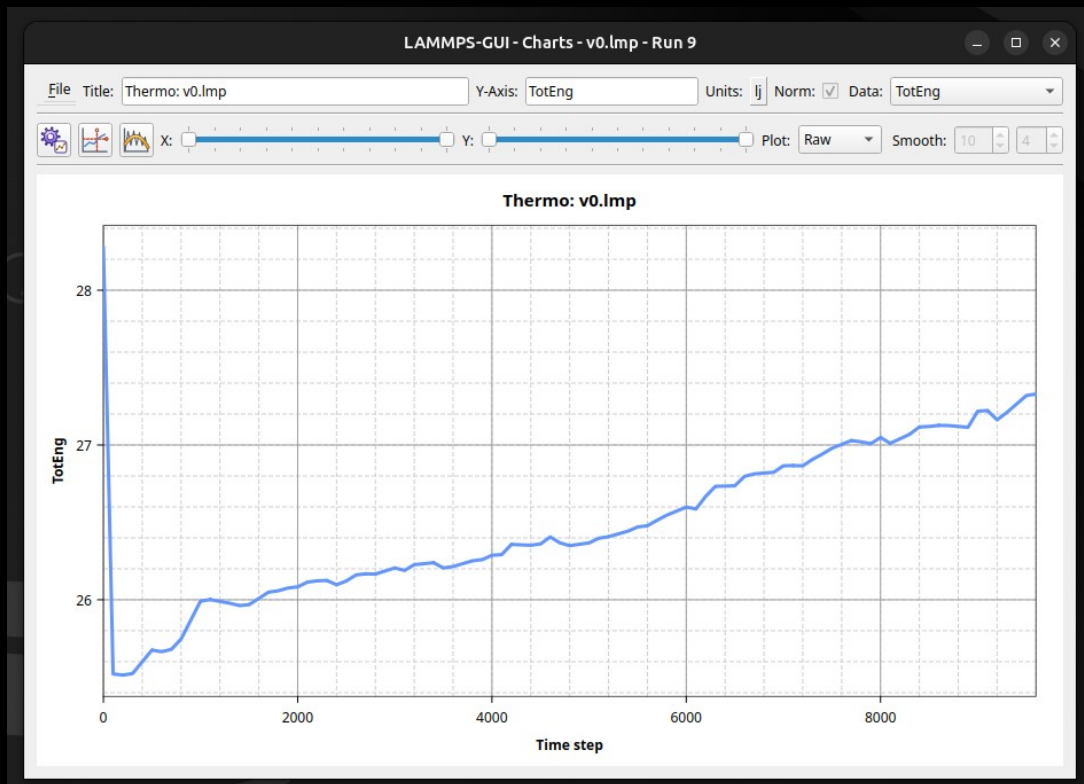
Lets modify the input

Increase the timestep to 0.005. What do you observe ?

```
15  
16 fix          mynve    all      nve # Integrate the equations of motion  
17 timestep     0.005 # Set the integration timestep to 0.005 in LJ reduced time units  
18
```

Increase the timestep to 0.005. What do you observe ?

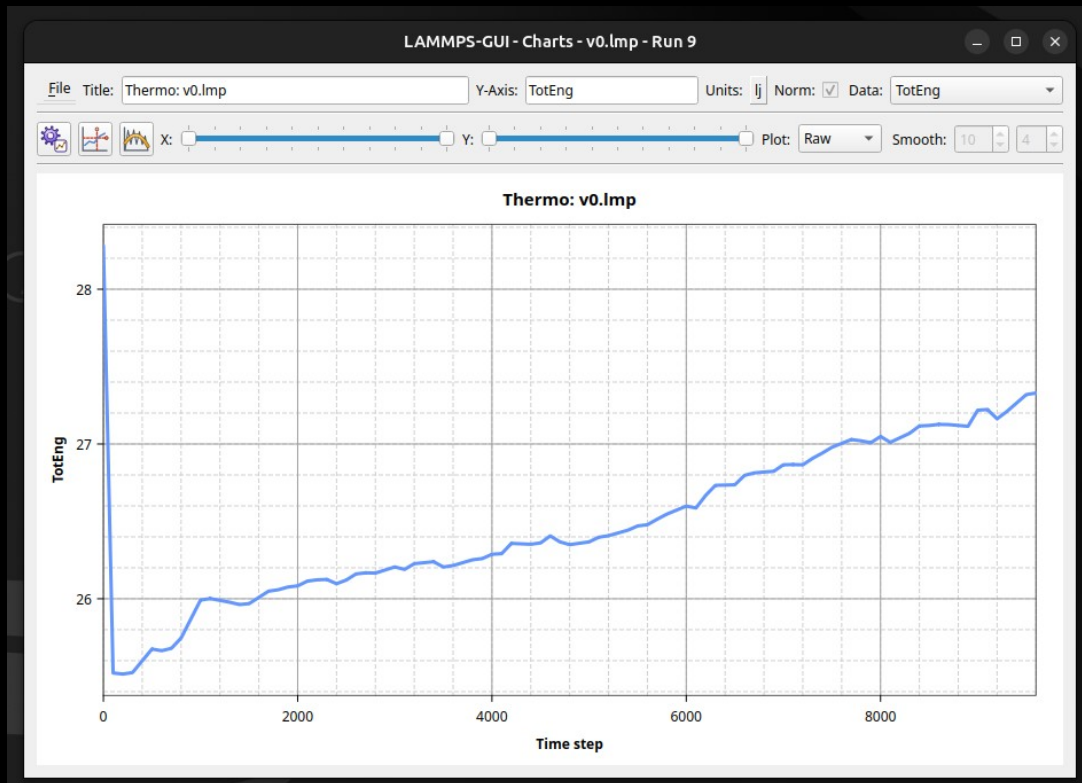
```
15  
16 fix          mynve      all          nve # Integrate the equations of motion  
17 timestep     0.005      # Set the integration timestep to 0.005 in LJ reduced time units  
18
```



There should be a drift in total energy.

Increase the timestep to 0.005. What do you observe ?

```
15  
16 fix          mynve      all          nve # Integrate the equations of motion  
17 timestep     0.005      # Set the integration timestep to 0.005 in LJ reduced time units  
18
```

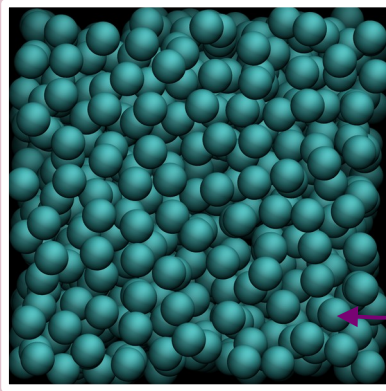


There should be a drift in total energy.

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

→ Accumulation of errors

Lets use a thermostat



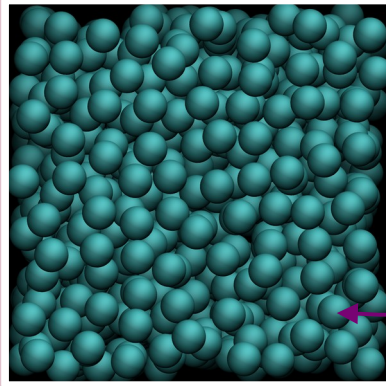
$$T_0 = 0.8$$

Energy

$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left( \frac{T_0}{T} - 1 \right)}$$

Lets use a thermostat



$$T_0 = 0.8$$

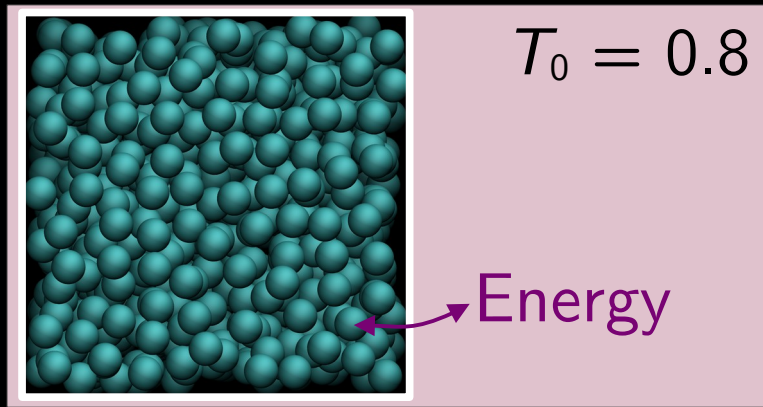
Energy

$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left( \frac{T_0}{T} - 1 \right)}$$

```
16 fix          mynve      all          nve # Integrate the equations of moti
17 timestep     0.0025 # Set the integration timestep to 0.0025 in LJ
18 fix          myber1     all          temp/berendsen 0.8 0.8 0.1 # Set the
19
```

Lets use a thermostat



$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

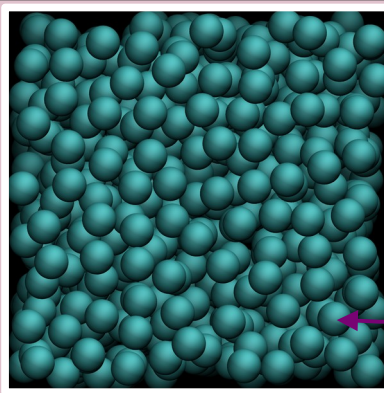
$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left( \frac{T_0}{T} - 1 \right)}$$

```
16 fix          mynve      all          nve # Integrate the equations of moti
17 timestep     0.0025 # Set the integration timestep to 0.0025 in LJ
18 fix          myber1     all          temp/berendsen 0.8 0.8 0.1 # Set the
```

Target temperature  $T_0$

Coupling time  $\tau$

Lets use a thermostat



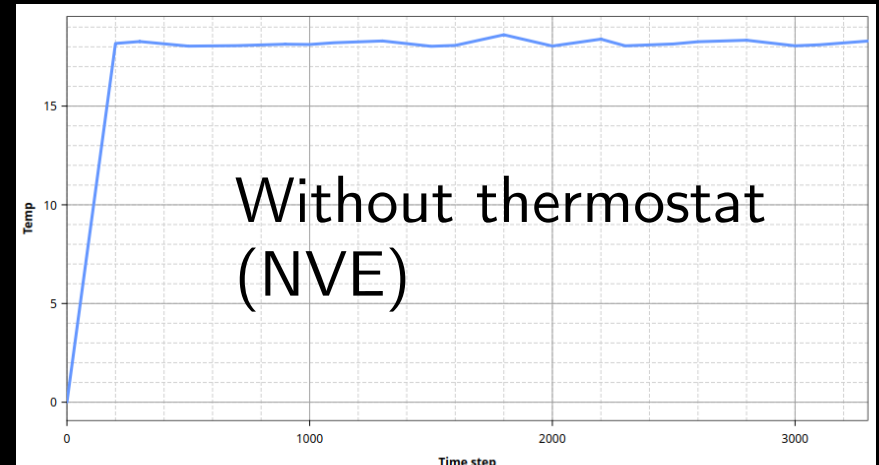
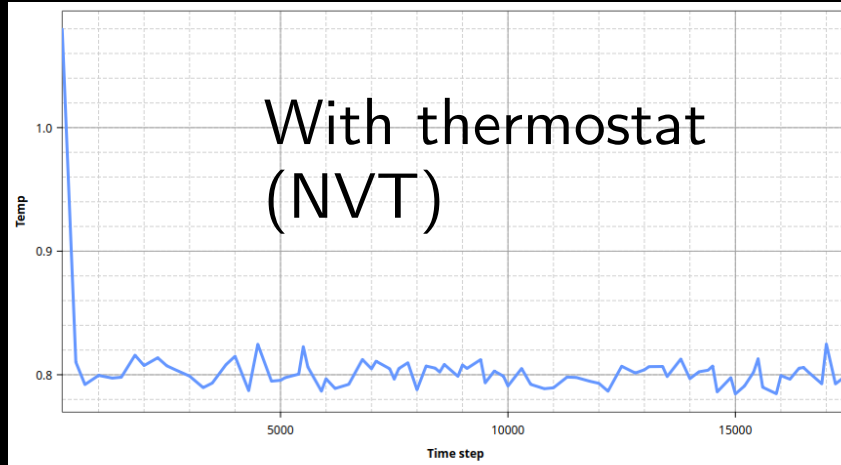
$$T_0 = 0.8$$

$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

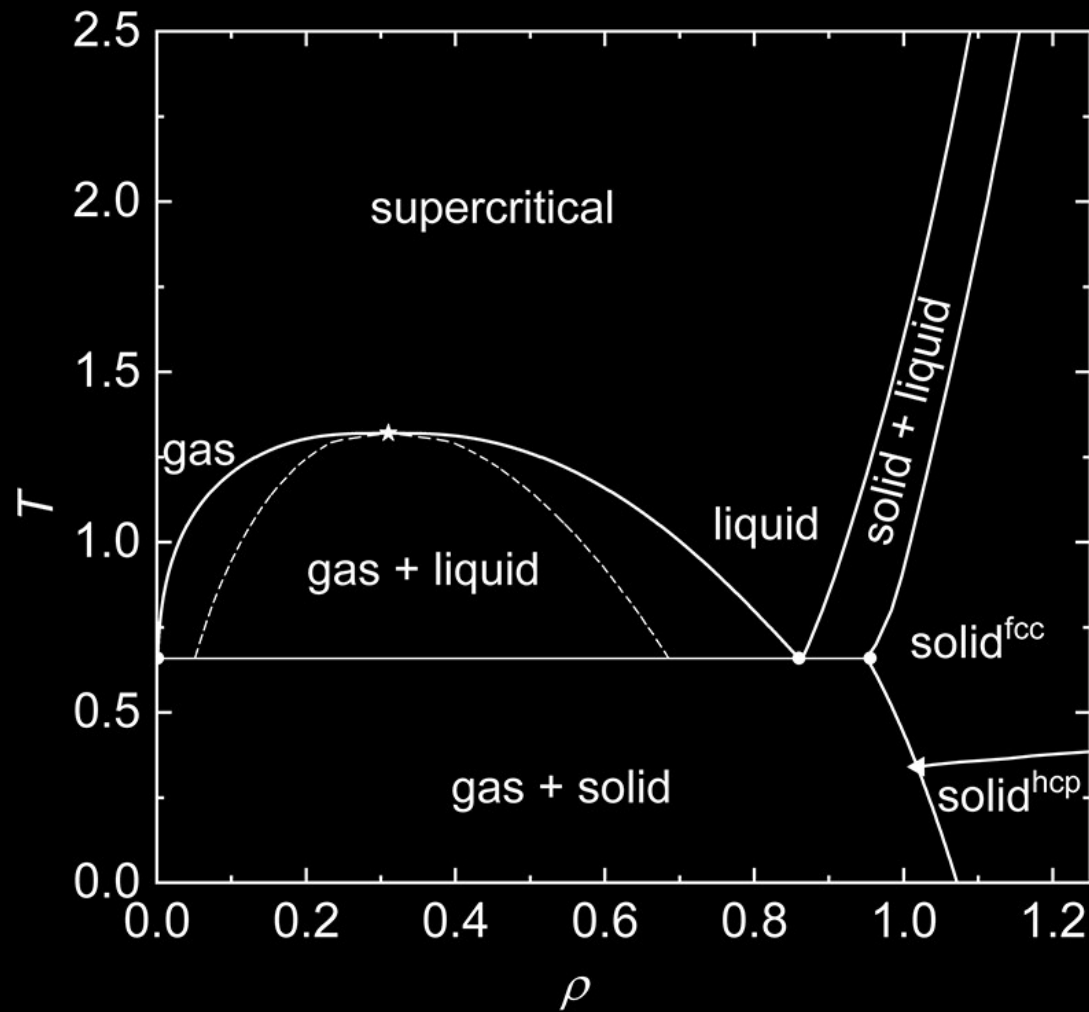
$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left( \frac{T_0}{T} - 1 \right)}$$

Energy

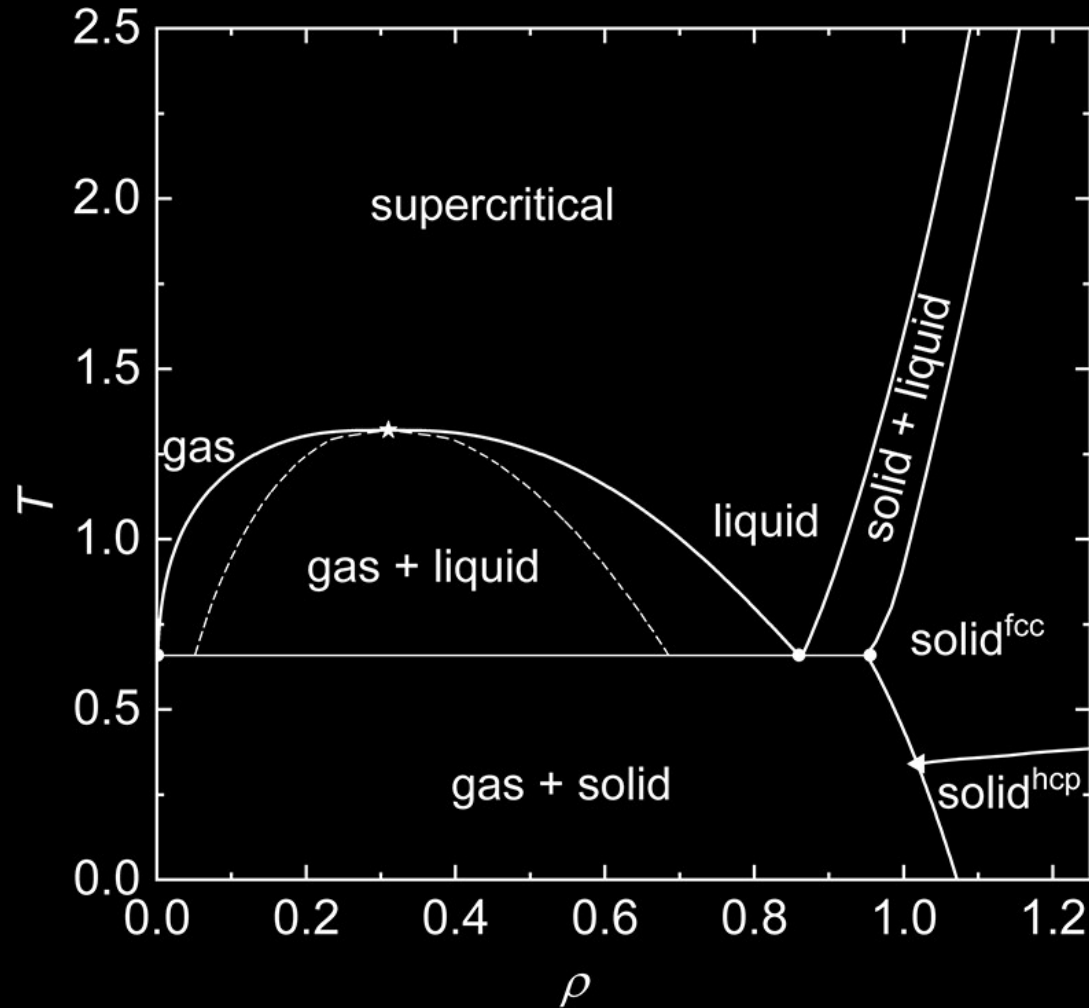
```
16 fix mynve all nve # Integrate the equations of moti
17 timestep 0.0025 # Set the integration timestep to 0.0025 in LJ
18 fix myber1 all temp/berendsen 0.8 0.8 0.1 # Set the
```



# Explore the phase diagram



## Explore the phase diagram

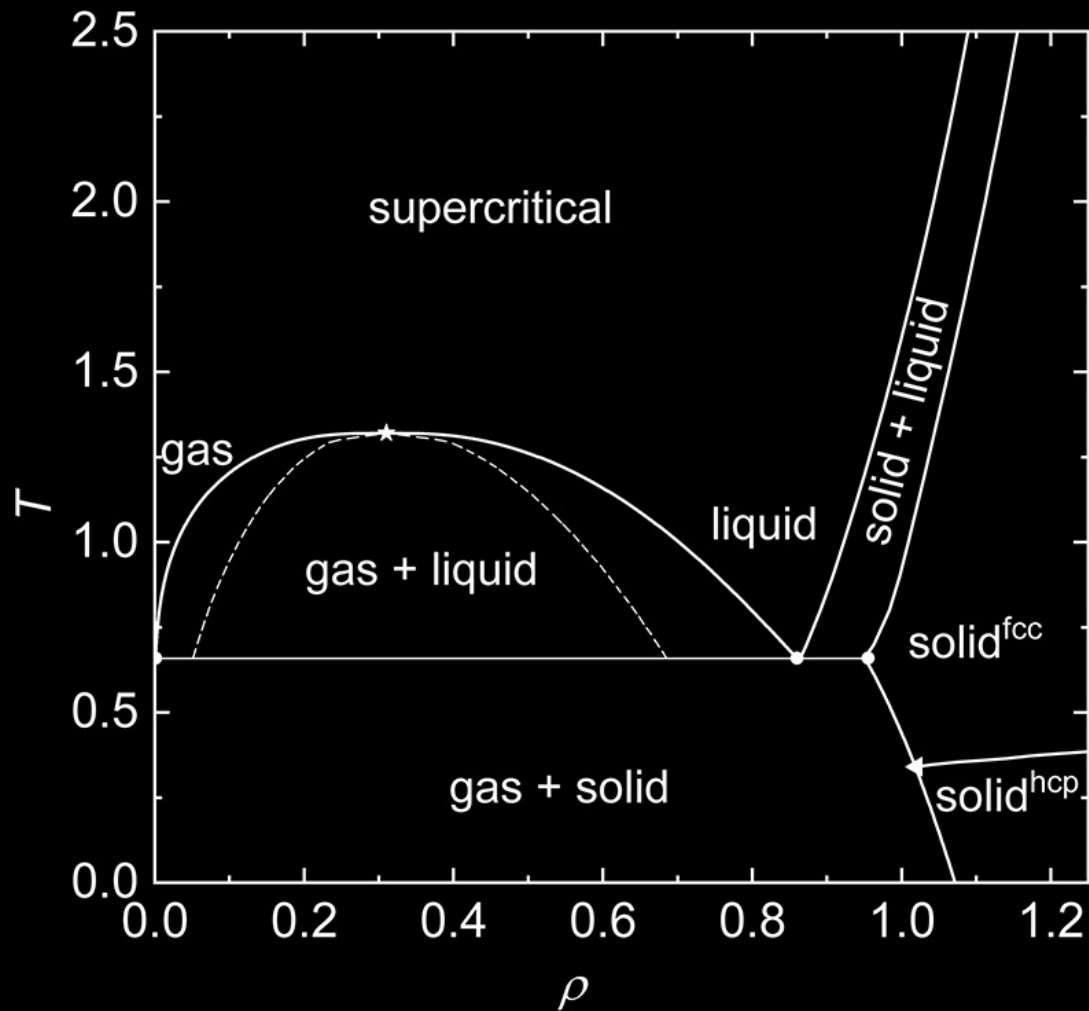


Current setup :

$$T = 0.8$$

$$\rho = 0.5$$

## Explore the phase diagram



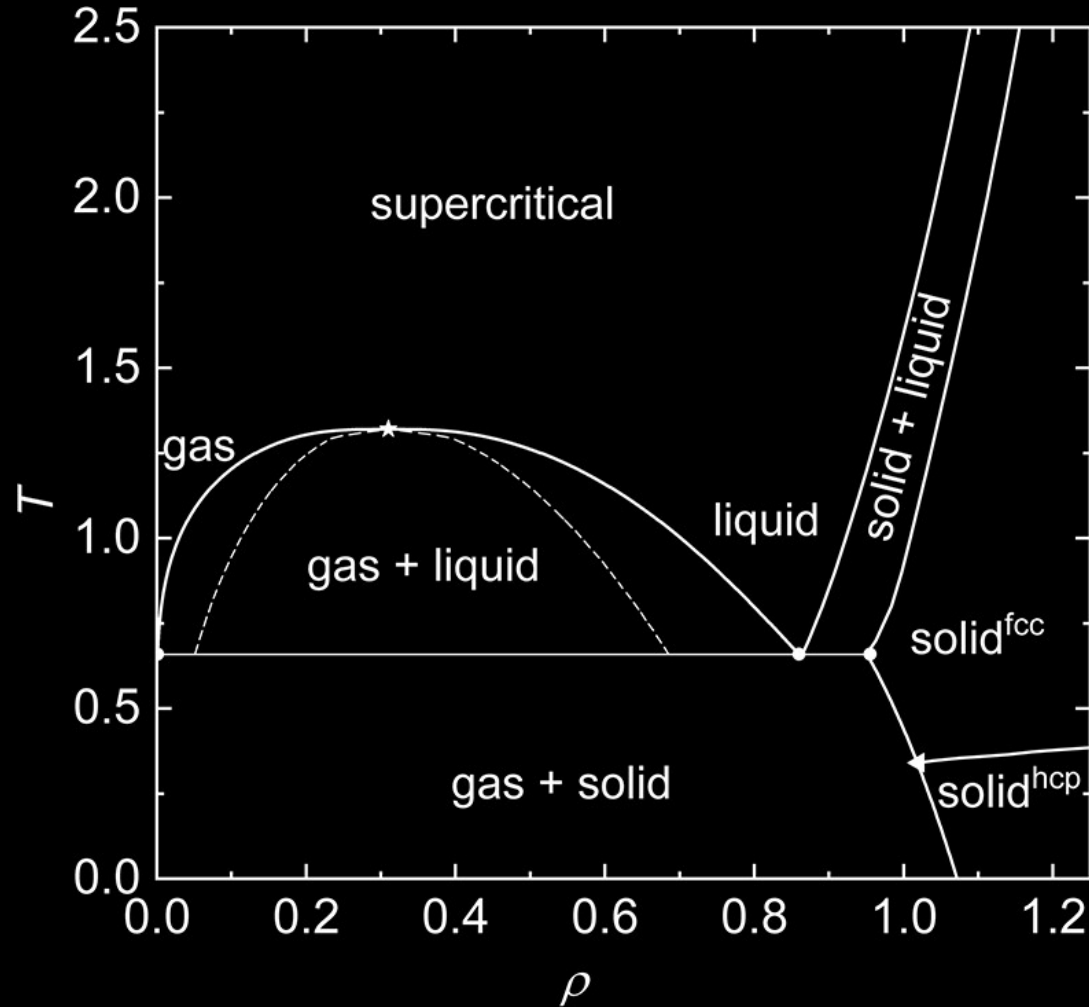
Current setup :

$$T = 0.8$$

$$\rho = 0.5$$

```
region      simbox block -6 6 -6 6 -6 6 #
create_box  1 simbox # Create a simulation box
create_atoms 1 random 864 34134 simbox over
```

## Explore the phase diagram

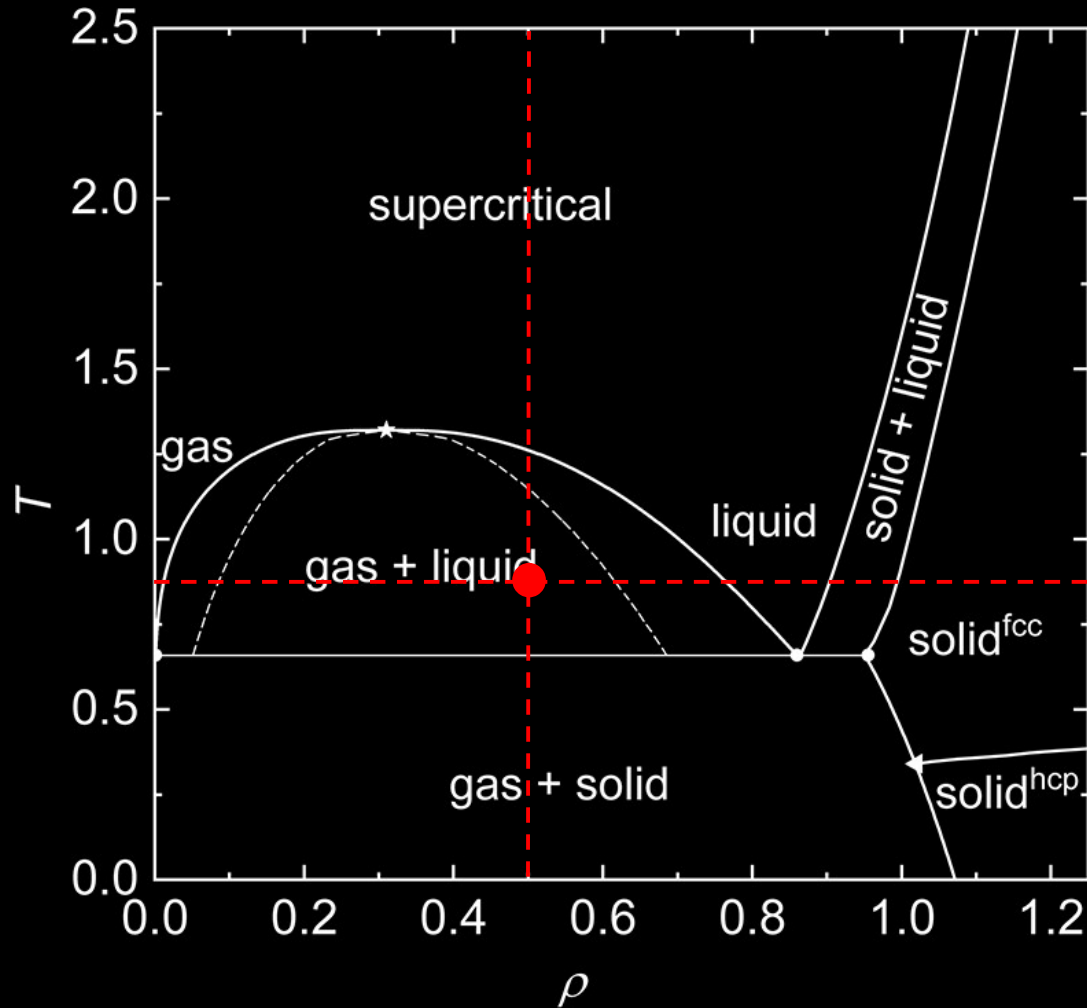


Current setup :

$$T = 0.8$$

$$\rho = 0.5$$

## Explore the phase diagram



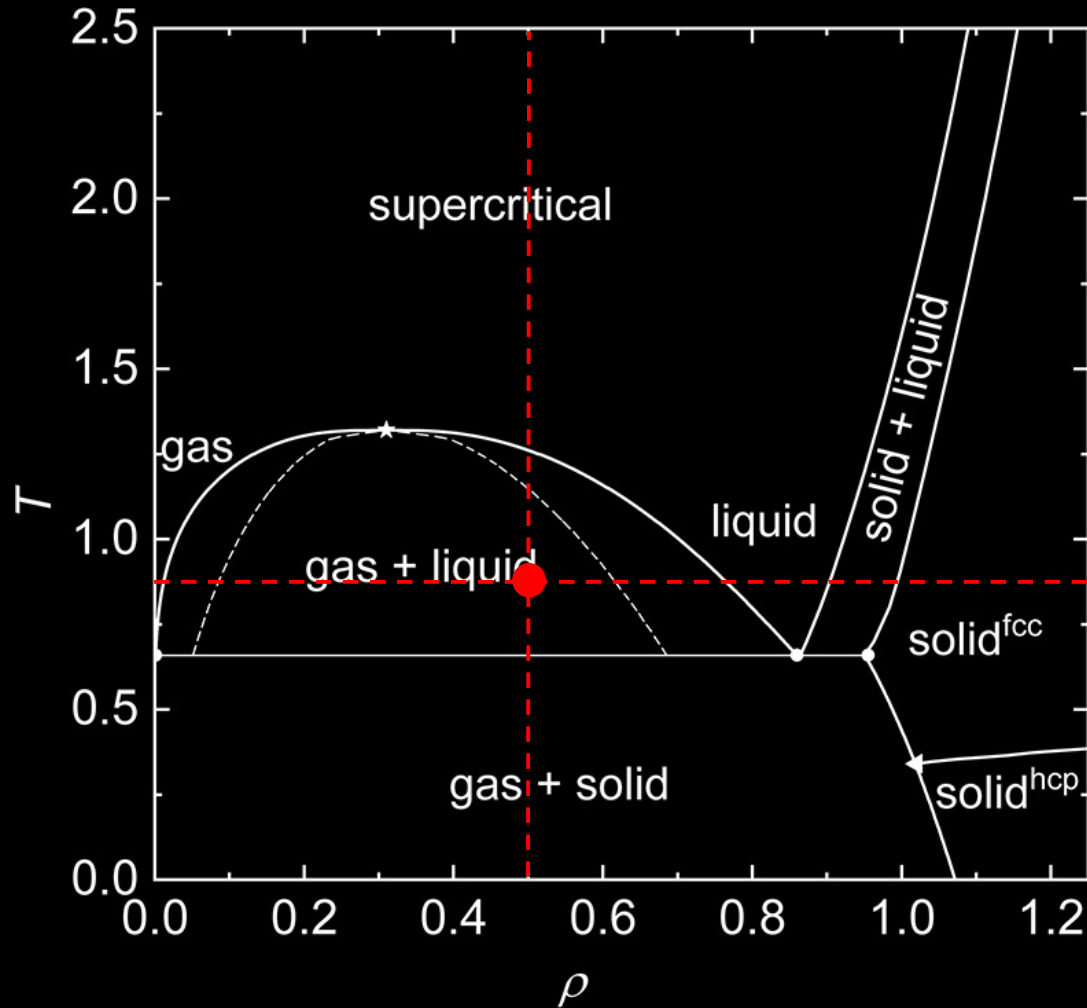
Current setup :

$$T = 0.8$$

$$\rho = 0.5$$

→ liquid gas coexistence

## Explore the phase diagram

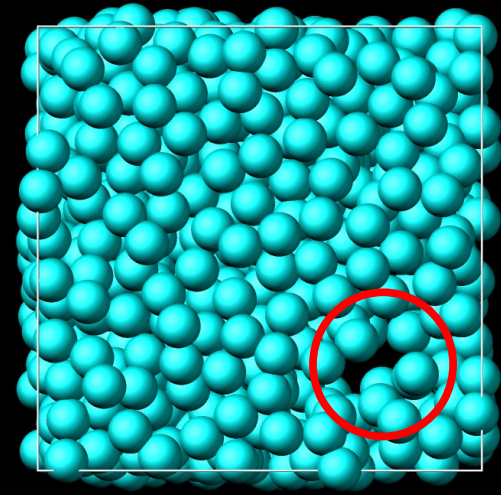


Current setup :

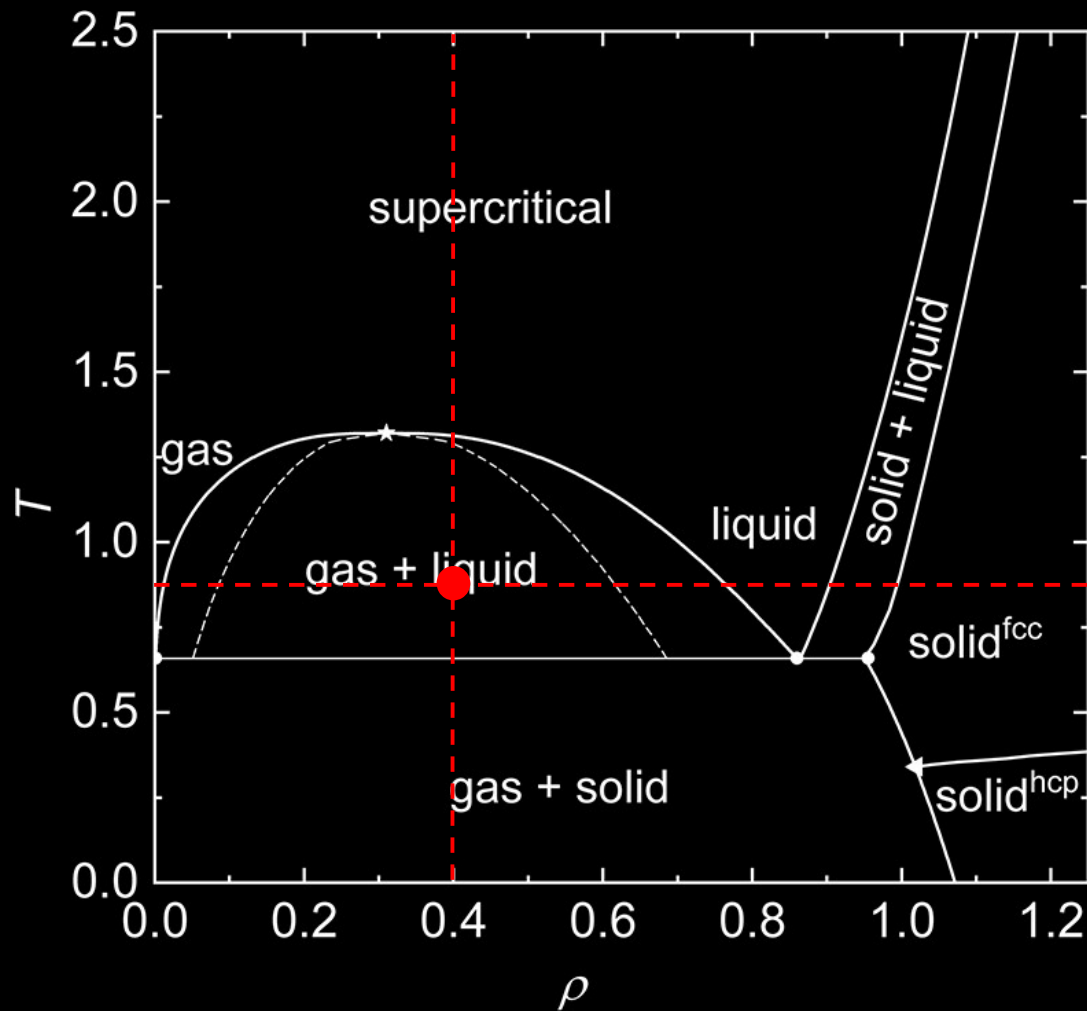
$$T = 0.8$$

$$\rho = 0.5$$

→ liquid gas coexistence



## Explore the phase diagram



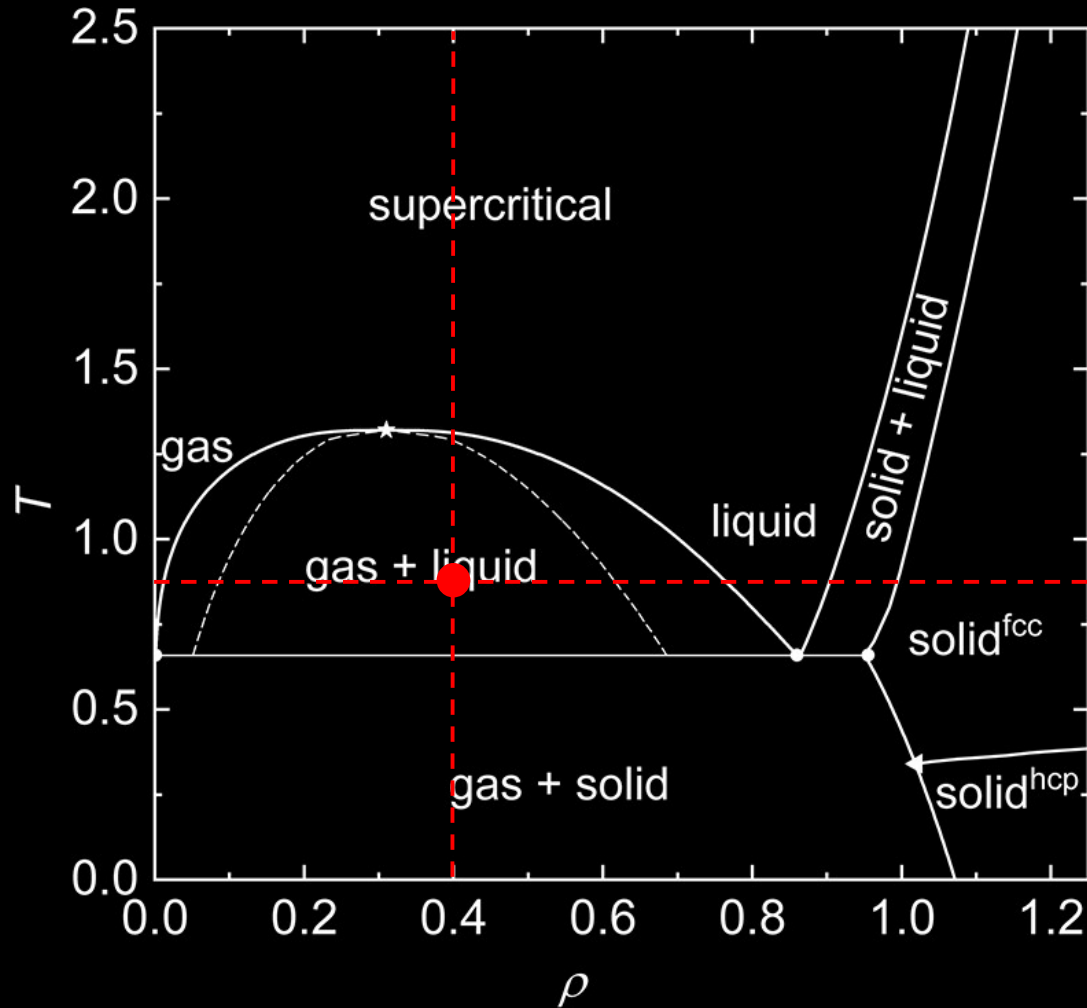
Reduce the density :

$$T = 0.8$$

$$\rho = 0.4 \text{ (use } N = 690 \text{)}$$

```
region      simbox block -6 6 -6 6 -6 6 # Define the
create_box  1 simbox # Create a simulation box containing
create_atoms 1 random 690 4134 simbox overlap 0.7 #
```

## Explore the phase diagram

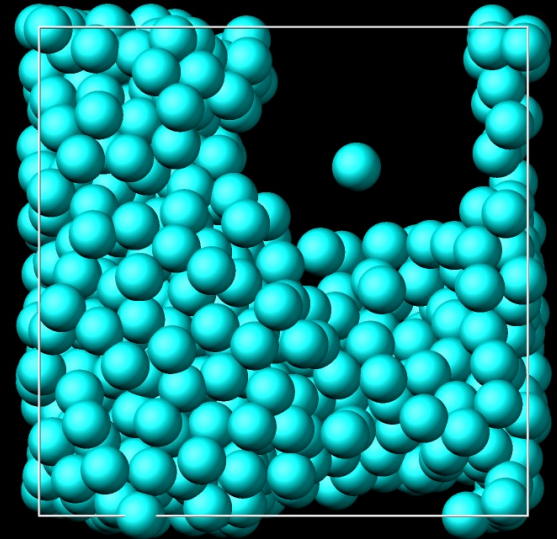


Reduce the density :

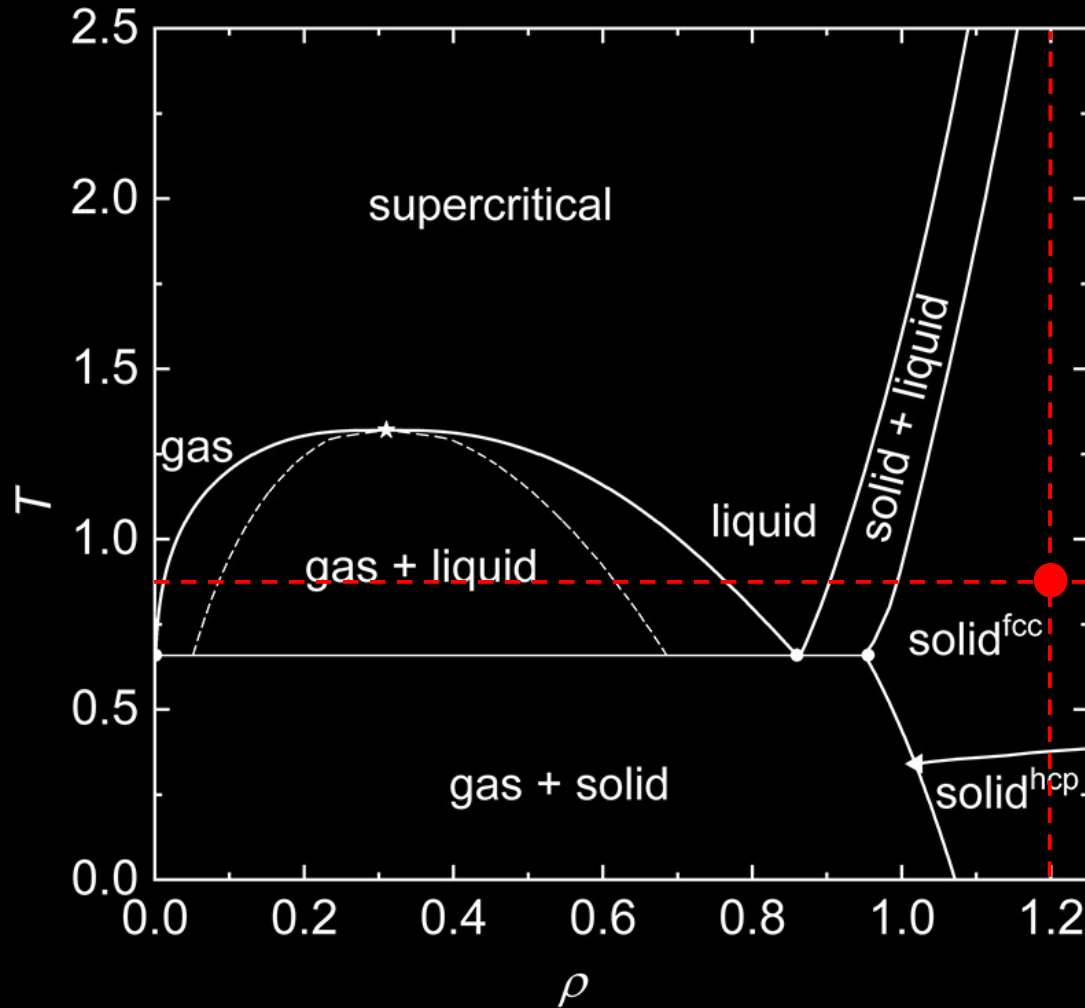
$$T = 0.8$$

$$\rho = 0.4 \text{ (use } N = 690 \text{)}$$

```
region      simbox block -6 6 -6 6 -6 6 # Define the  
create_box  1 simbox # Create a simulation box conta  
create_atoms 1 random 690 4134 simbox overlap 0.7 #
```



## Explore the phase diagram



Simulate a solid :

$$T = 0.8$$

$$\rho = 1.2 \text{ (use } L = 9 \text{)}$$

```
region          simbox block -4.5 4.5  
create_box      1 simbox # Create a si  
create_atoms    1 random 864 34134 sim
```

Explore the phase diagram

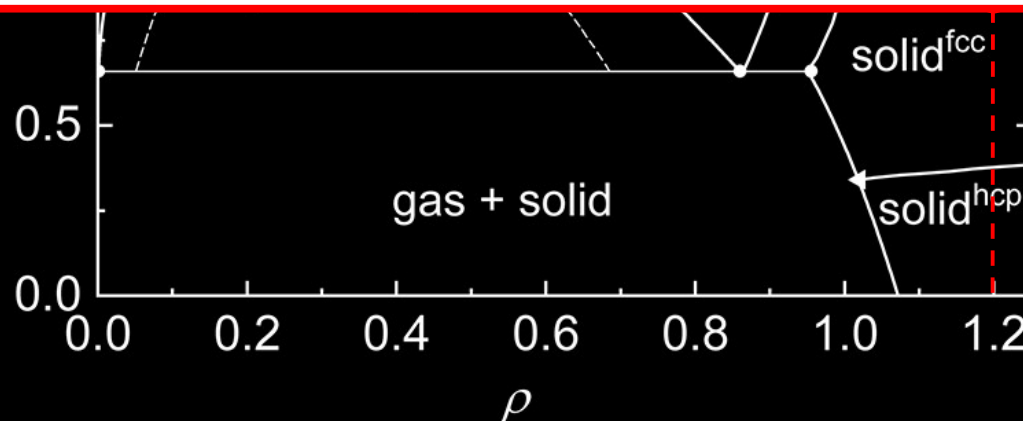


Simulate a solid :

$$T = 0.8$$

$$\rho = 1.2 \text{ (use } L = 9 \text{)}$$

```
region          simbox block -4.5 4.5 -4.5 4.5 -4.5 4.5 #  
create_box      1 simbox # Create a simulation box conta  
create_atoms    1 random 864 34134 simbox overlap 0.7 # P
```



Explore the phase diagram

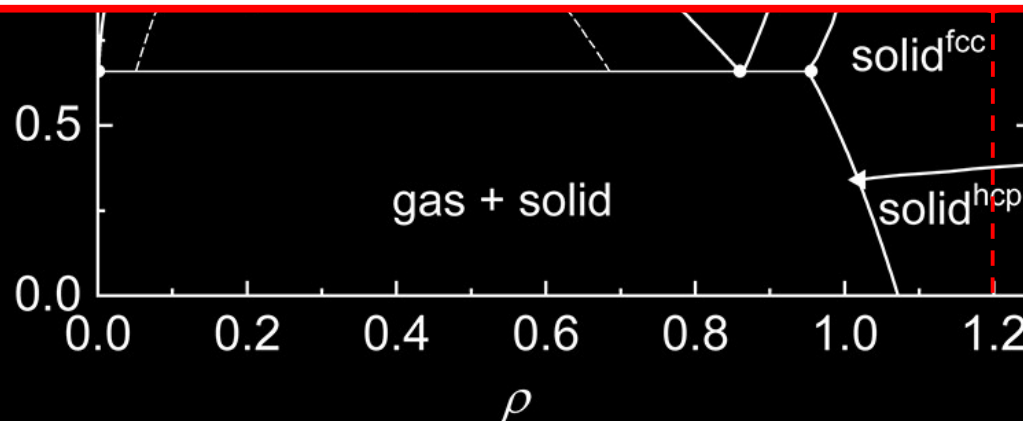


Simulate a solid :

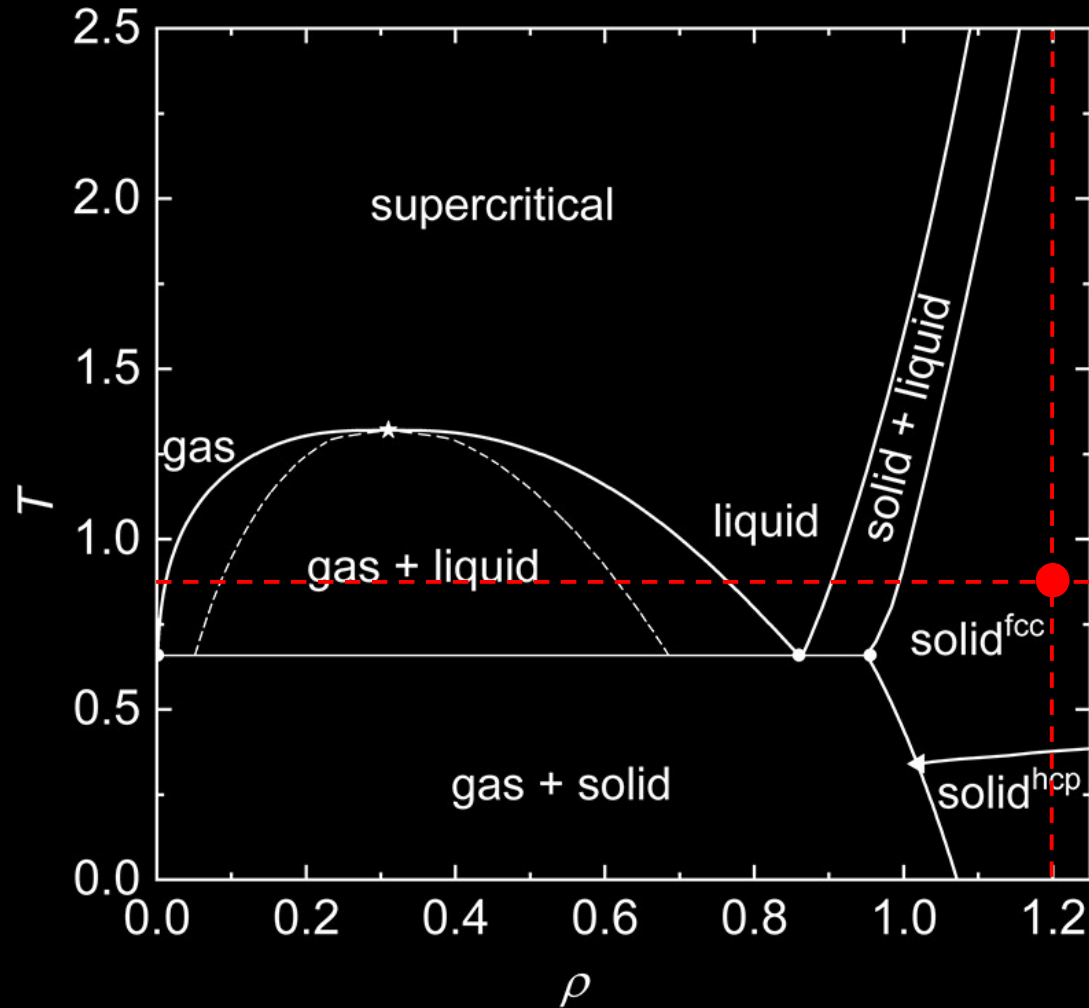
$$T = 0.8$$

$$\rho = 1.2 \text{ (use } L = 9\text{)}$$

```
region                                simbox block -4.5 4.5 -4.5 4.5 -4.5 4.5 #  
create_box                           1 simbox # Create a simulation box conta  
create_atoms                         1 random 864 34134 simbox overlap 0.7 # P
```



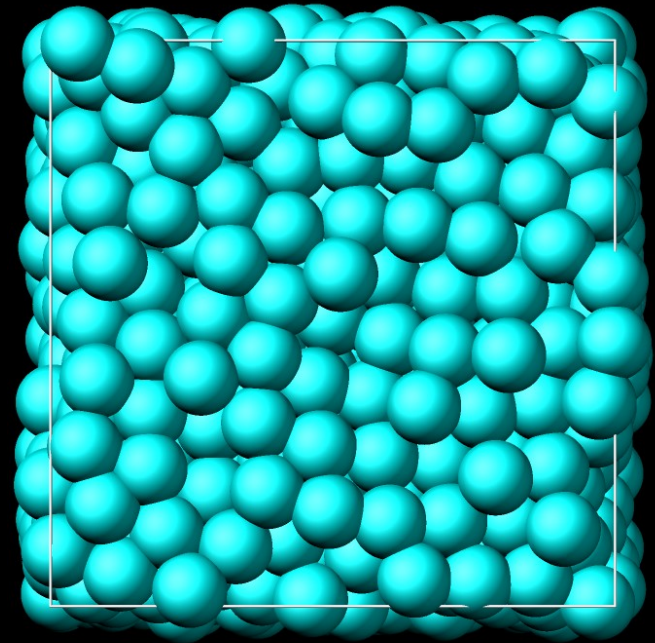
## Explore the phase diagram



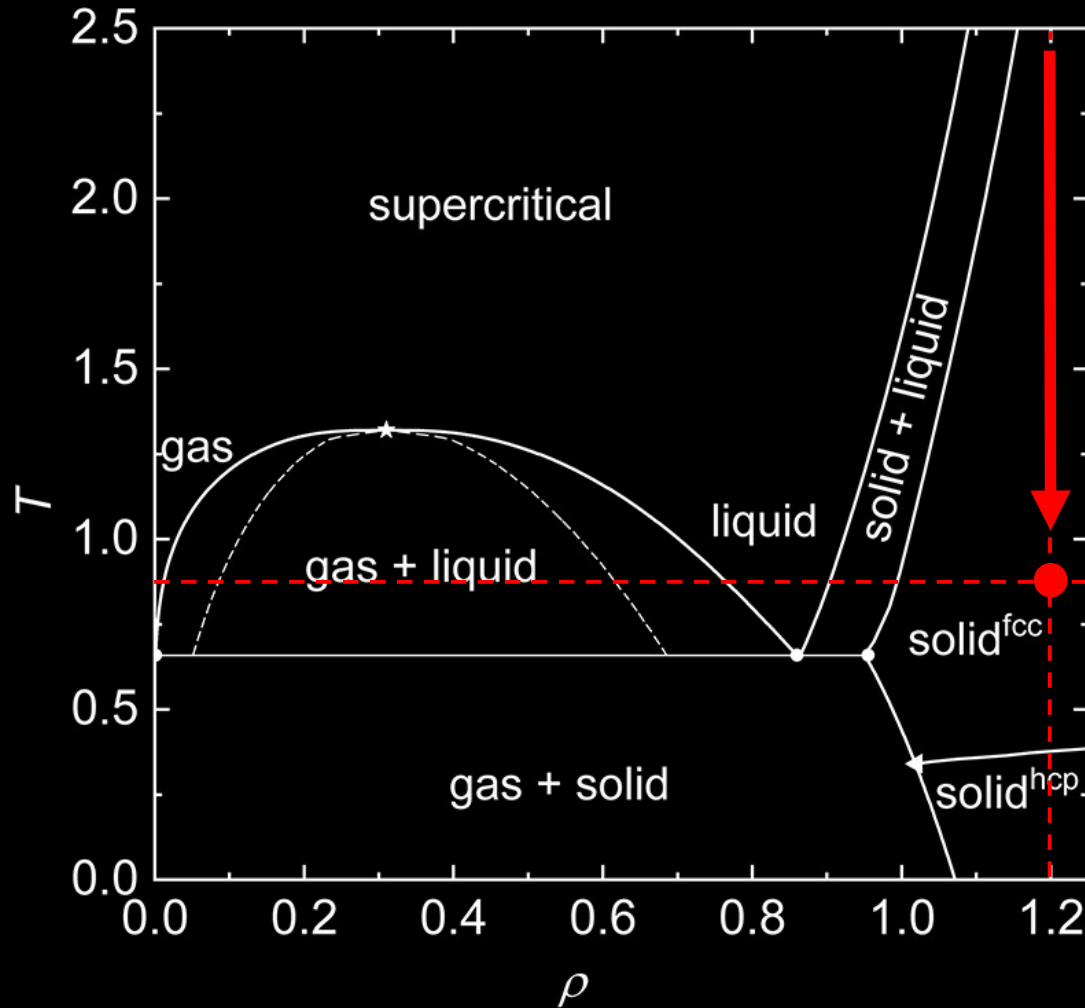
Simulate a solid :

$$T = 0.8$$

$$\rho = 1.2 \text{ (use } L = 9 \text{)}$$



## Explore the phase diagram



Slow temperature ramp

From  $T = 5$  to 0.8

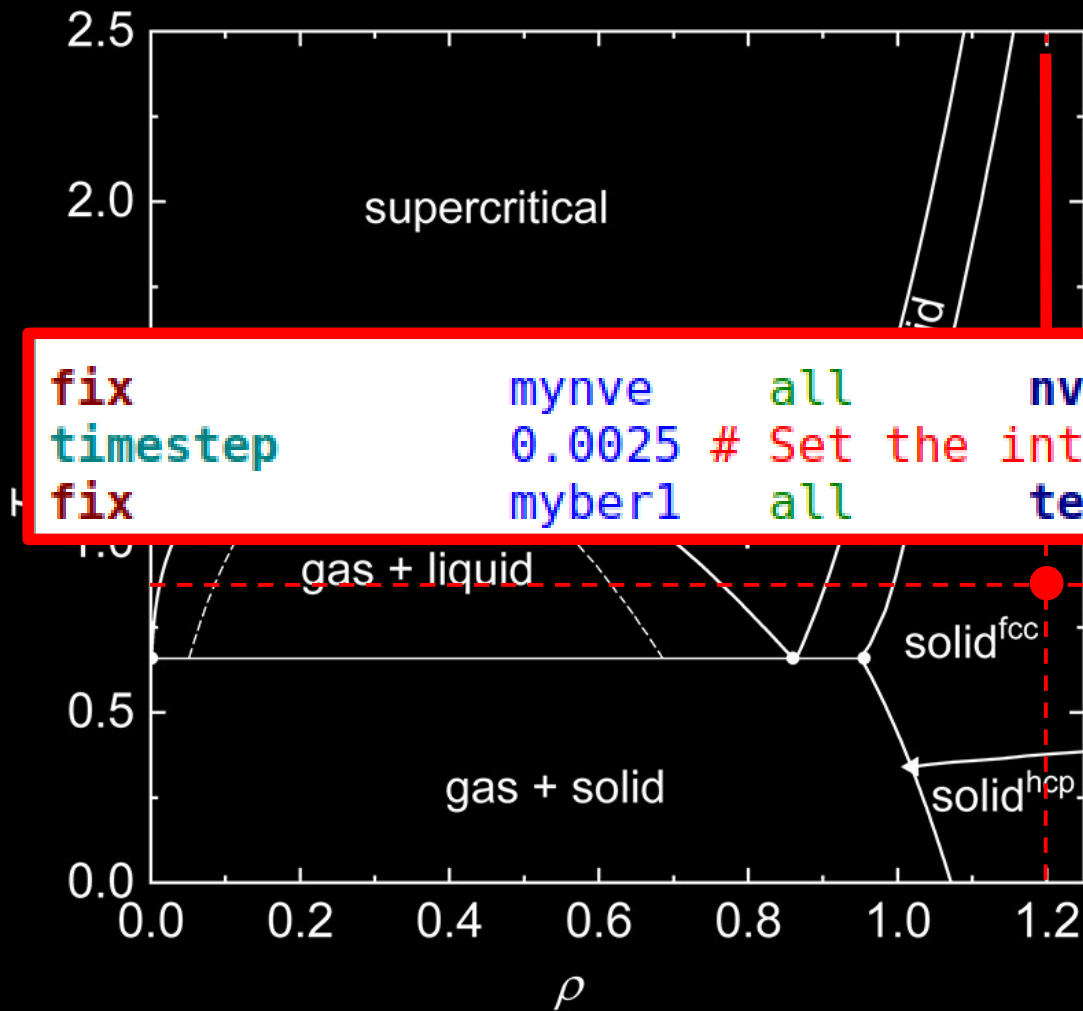
$\rho = 1.2$

## Explore the phase diagram

Slow temperature ramp

From  $T = 5$  to  $0.8$

$\rho = 1.2$

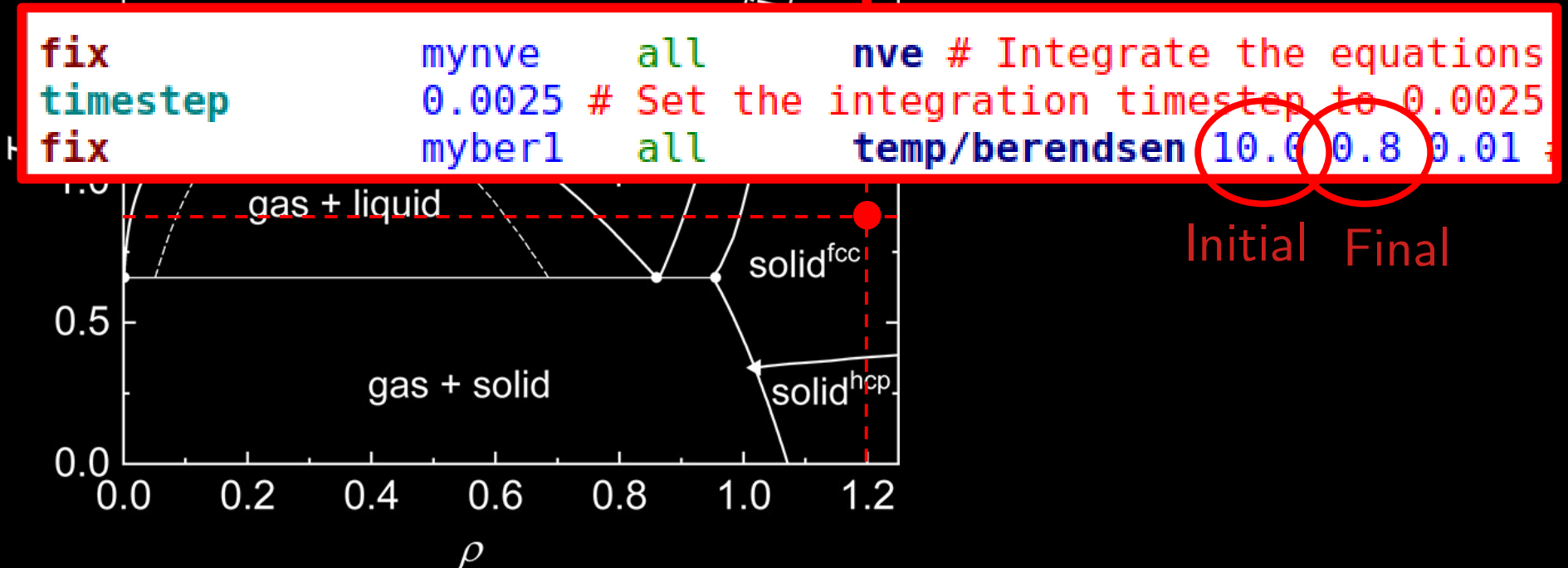


## Explore the phase diagram

Slow temperature ramp

From  $T = 5$  to 0.8

$\rho = 1.2$

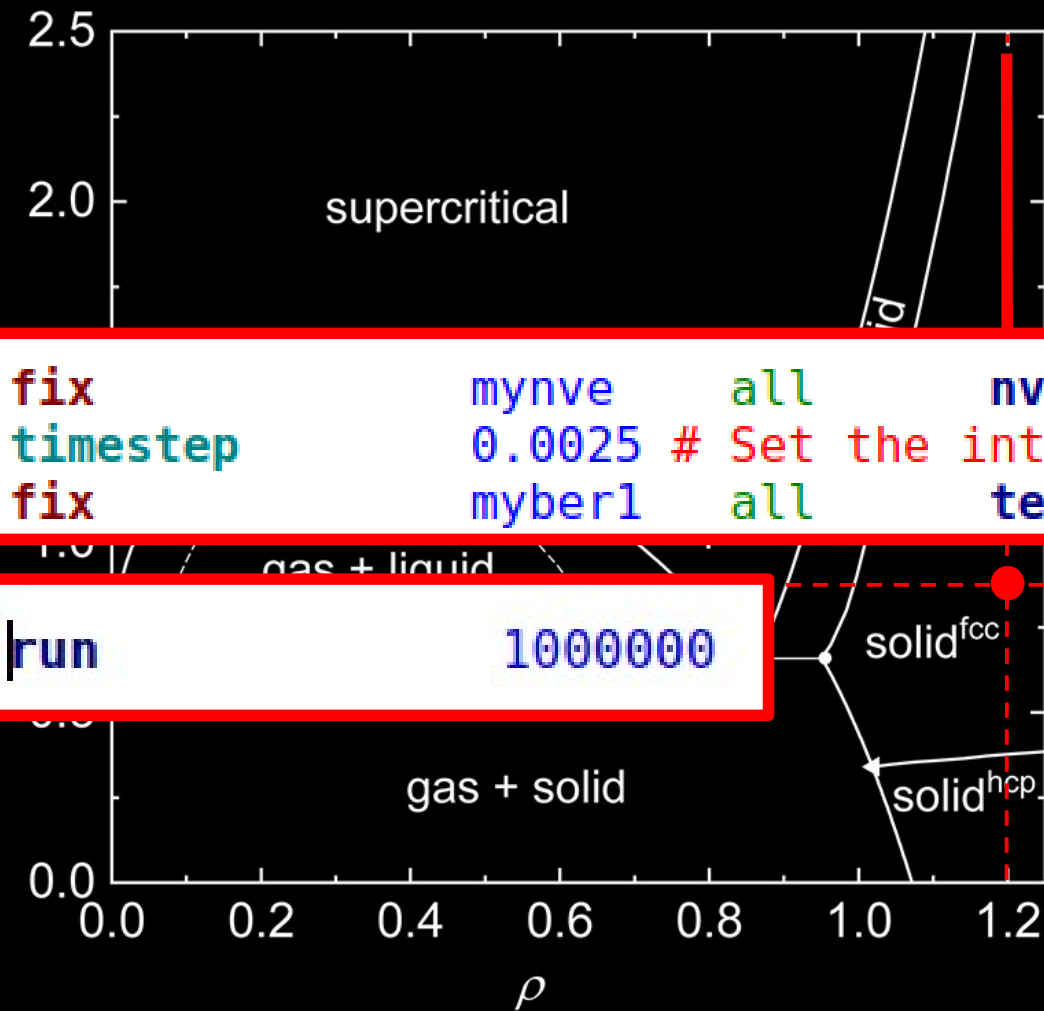


## Explore the phase diagram

Slow temperature ramp

From  $T = 5$  to  $0.8$

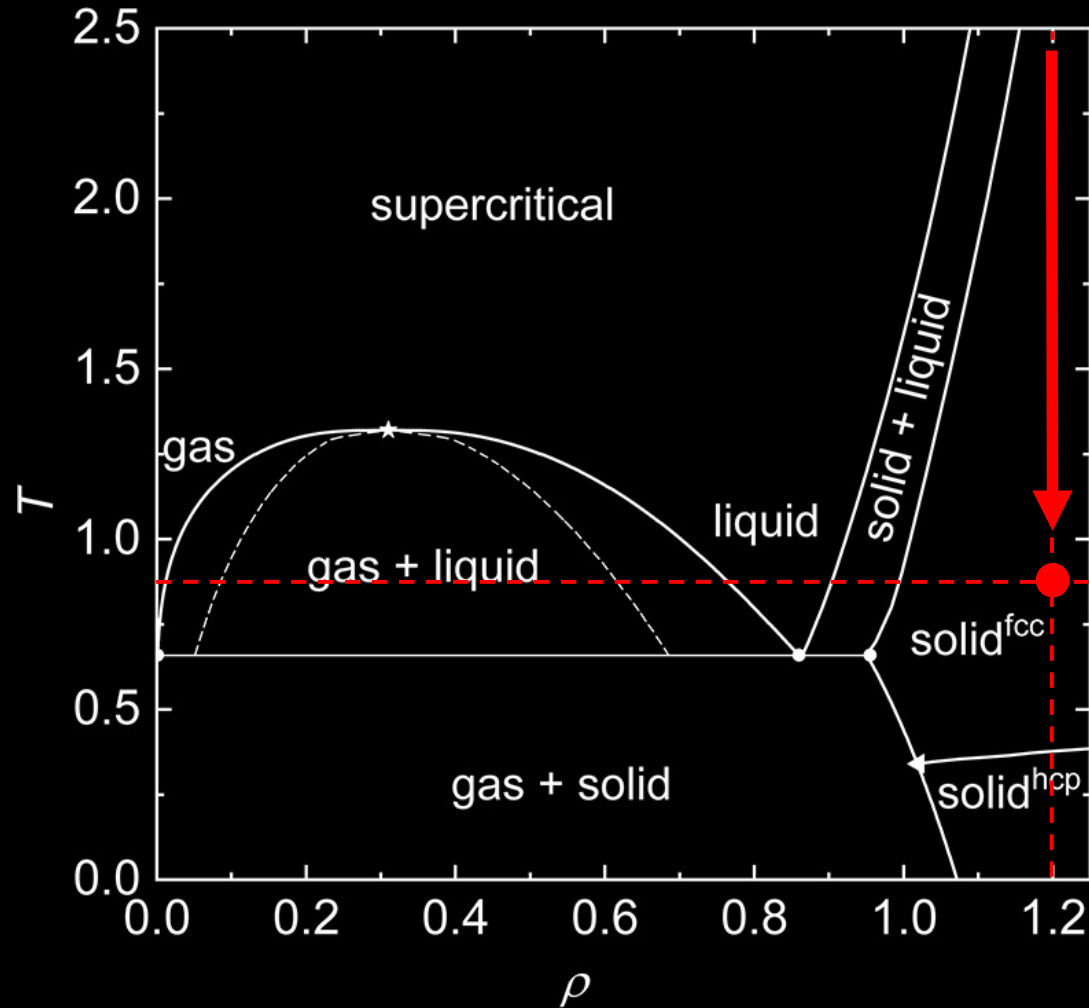
$\rho = 1.2$



```
fix          mynve      all          nve # Integrate the equations
timestep     0.0025 # Set the integration timestep to 0.0025
fix          myber1     all          temp/berendsen 10.0 0.8 0.01
```

```
run          1000000
```

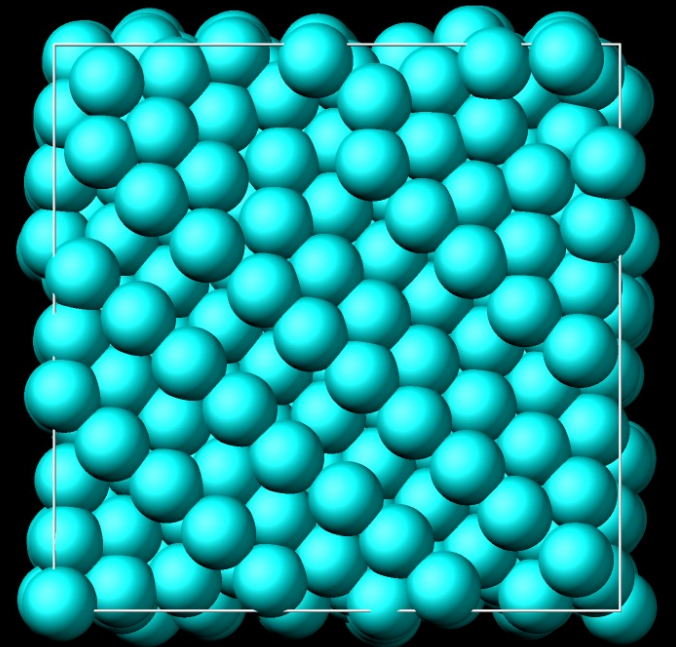
## Explore the phase diagram



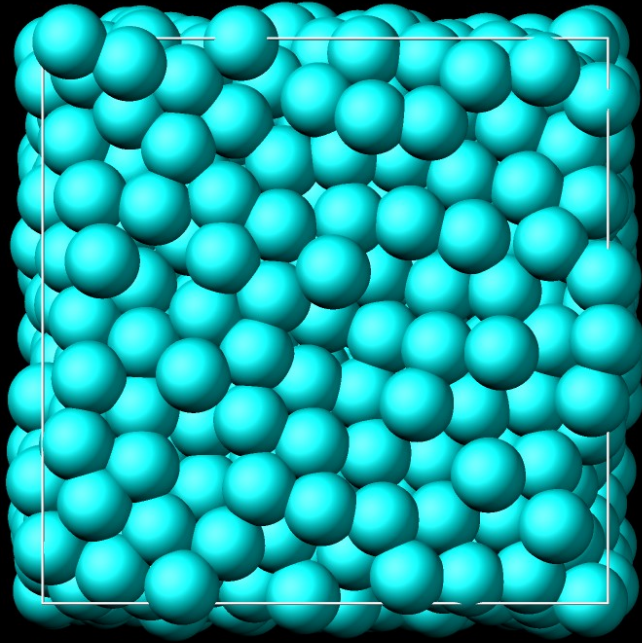
## Slow temperature ramp

From  $T = 5$  to 0.8

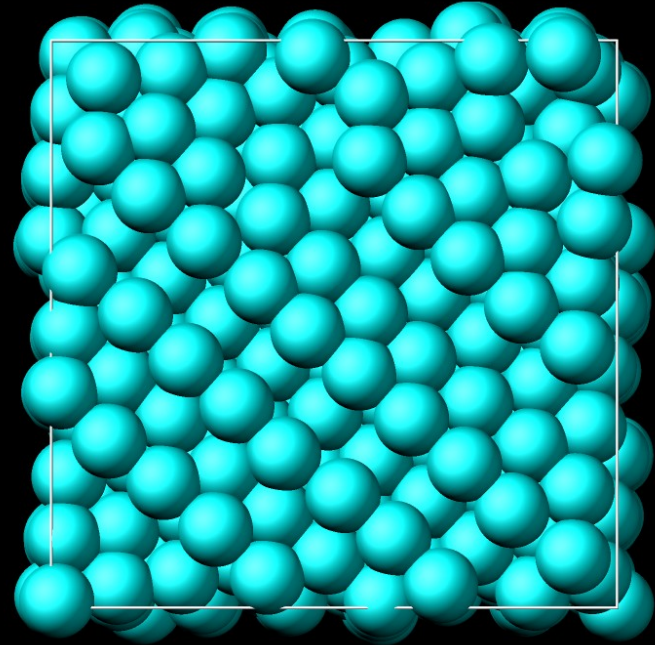
$\rho = 1.2$



Fast quenching



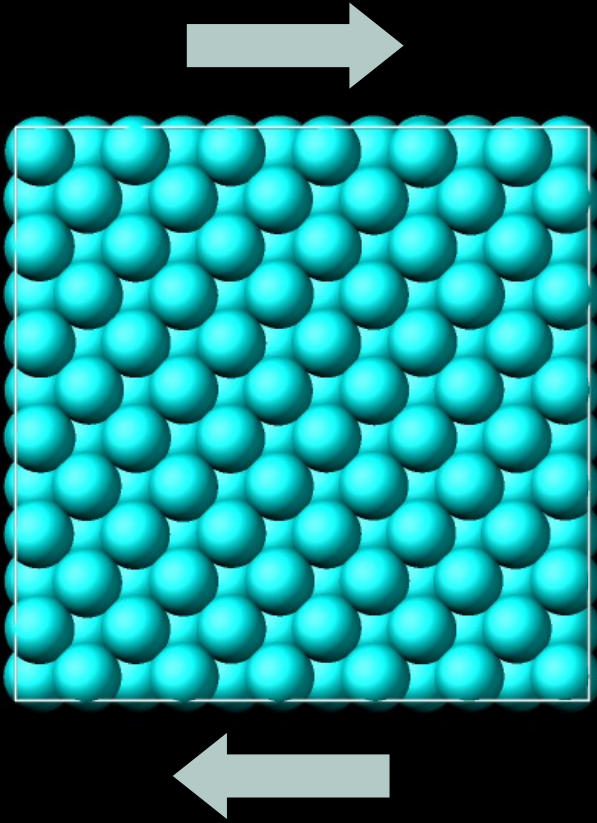
“Slow” temperature ramp



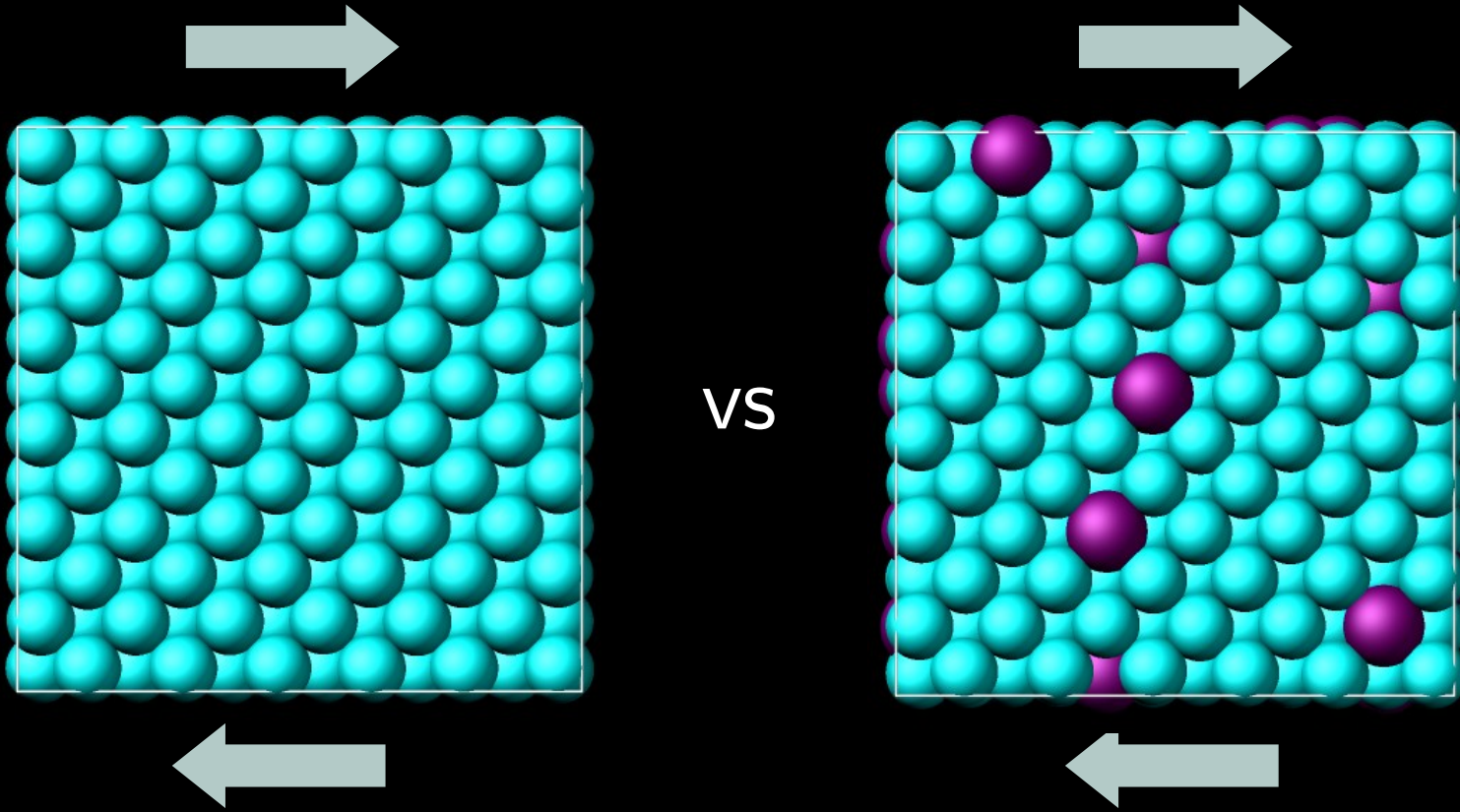
Going further

# Effect of impurity on the stress-strain response

# Effect of impurity on the stress-strain response



# Effect of impurity on the stress-strain response



# <https://github.com/simongravelle/PHELMA>

## Second input

```
# Simple NVT simulation of a fcc crystal with impurities

units          lj # Use Lennard-Jones reduced units
dimension      3 # Perform the simulation in 3 spatial dimensions
atom_style     atomic # Atoms are treated as point particles without bonds or molecular topology
boundary       p p p # Use periodic boundary conditions

lattice        fcc 1.55 origin 0.25 0.25 0.25
region         simbox prism -3 3 -3 3 -3 3 0.0 0.0 0.0 # Define the simulation box from -6 to 6 in x, y, z
create_box     2 simbox # Create a simulation box containing two atom type
create_atoms   1 region simbox # Create atoms along the predefined lattice
lattice        none 1

mass           1 1.0 # Assign a mass to type one
mass           2 1.7 # Assign a mass to type one
pair_style     lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
pair_coeff     1 1 1.0 1.0 # Set Lennard-Jones parameters for atom type 1
pair_coeff     2 2 1.0 1.2 # Set Lennard-Jones parameters for atom type 2 (atoms of type are slightly different)

set            group all type/fraction 2 0.0 46415 # Replace a fraction X of atoms with type 2

velocity       all create 0.1 4928459 rot yes mom yes dist gaussian
fix            mynve all nve # Integrate the equations of motion
timestep       0.0025 # Set the integration timestep to 0.0025 in LJ reduced time units
fix            myber1 all temp/berendsen 0.1 0.1 0.1 # Set the temperature (unitless)

thermo         100 # Print thermodynamic information in log
thermo_style   custom step temp etotal ke pe density pxy pyz pxz # Choose what information is printed

dump           viz all image 100 myimage-*.ppm type type size 800 800 zoom 1.00 shiny 0.5 fsaa y
dump_modify    viz pad 9 boxcolor white backcolor black adiam 1 1 adiam 2 1.2 acolor 1 cyan acolor 2 purple

run            1000 # System equilibration
```

```
lattice      fcc 1.55 origin 0.25 0.25 0.25
region      simbox prism -3 3 -3 3 -3 3 0.0 0.0 0.0 # D
create_box   2 simbox # Create a simulation box containing
create_atoms 1 region simbox # Create atoms along the pr
lattice      none 1
```

```
lattice      fcc 1.55 origin 0.25 0.25 0.25
region      simbox prism -3 3 -3 3 -3 3 0.0 0.0 0.0 # D
create_box  2 simbox # Create a simulation box containing
create_atoms 1 region simbox # Create atoms along the pr
lattice      none 1
```

- Atoms are placed on a lattice

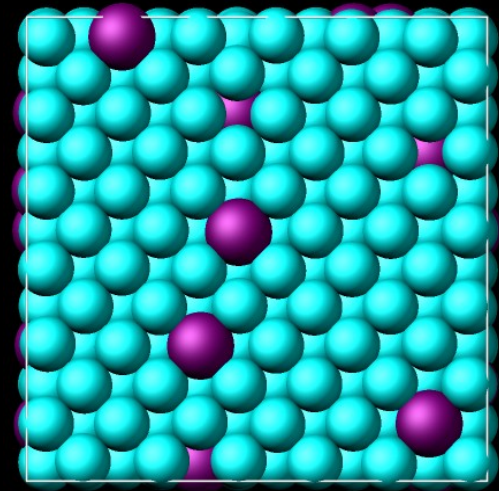
```
lattice          fcc 1.55 origin 0.25 0.25 0.25
region          simbox prism -3 3 -3 3 -3 3 0.0 0.0 0.0 # D
create_box      2 simbox # Create a simulation box containing
create_atoms    1 region simbox # Create atoms along the pr
lattice         none 1
```

- Atoms are placed on a lattice
- The box contains two types of atoms

```
mass      1      1.0 # Assign a
mass      2      1.7 # Assign a
pair_style lj/cut 4.0 # Use a l
pair_coeff 1      1      1.0 1.0 #
pair_coeff 2      2      1.0 1.2 #
```

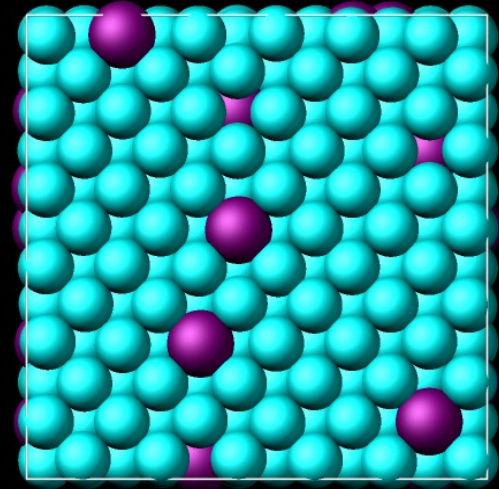
- Parameters are set for 2 types of atoms

```
mass      1      1.0 # Assign a
mass      2      1.7 # Assign a
pair_style lj/cut 4.0 # Use a
pair_coeff 1      1      1.0 1.0 #
pair_coeff 2      2      1.0 1.2 #
```



- Parameters are set for 2 types of atoms

```
mass      1      1.0 # Assign a
mass      2      1.7 # Assign a
pair_style lj/cut 4.0 # Use a
pair_coeff 1      1      1.0 1.0 #
pair_coeff 2      2      1.0 1.2 #
```



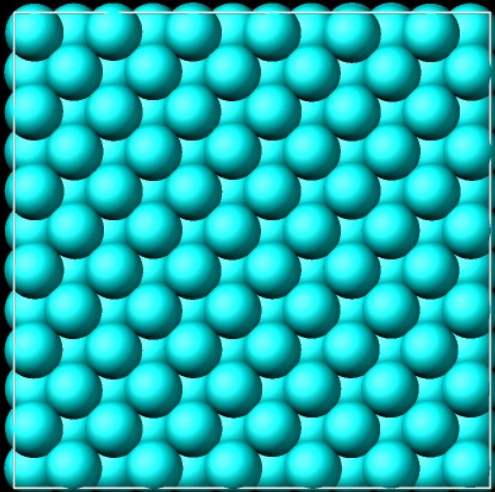
Atom type “2” is larger

- Parameters are set for 2 types of atoms

```
set          group all type/fraction 2 0.0 46415 #
```

## Fraction of atoms to replace

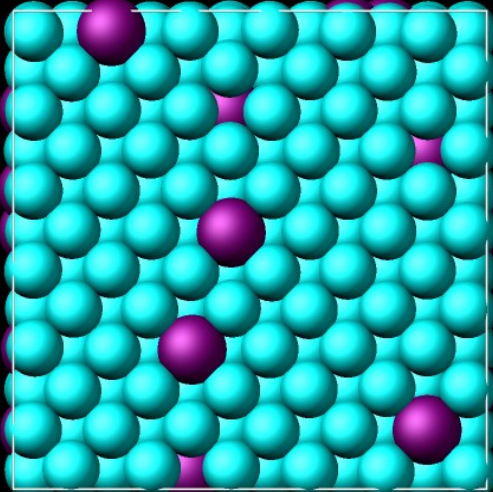
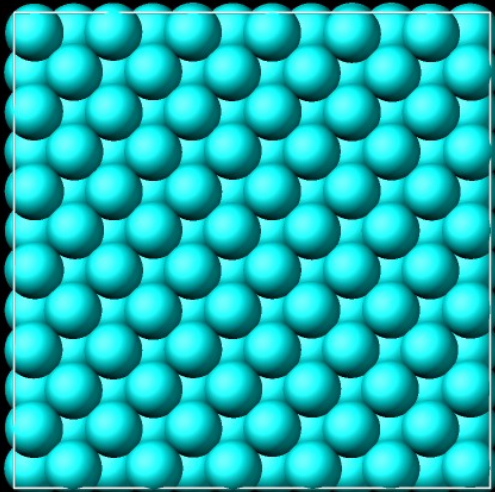
```
set          group all type/fraction 2 0.0 46415 #
```



## Fraction of atoms to replace

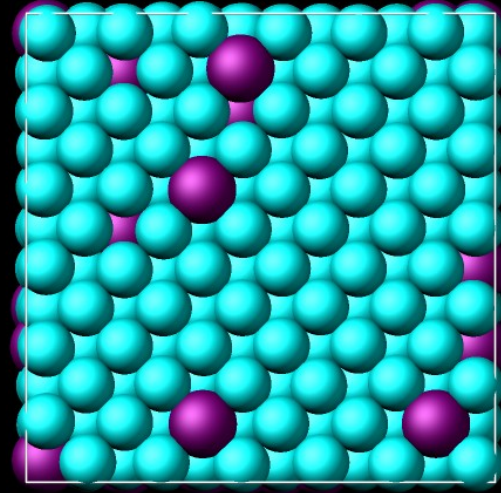
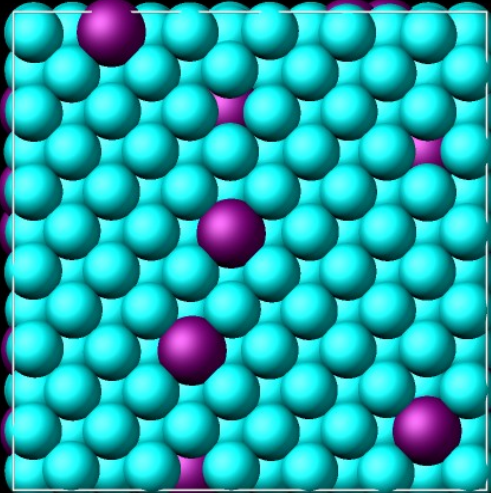
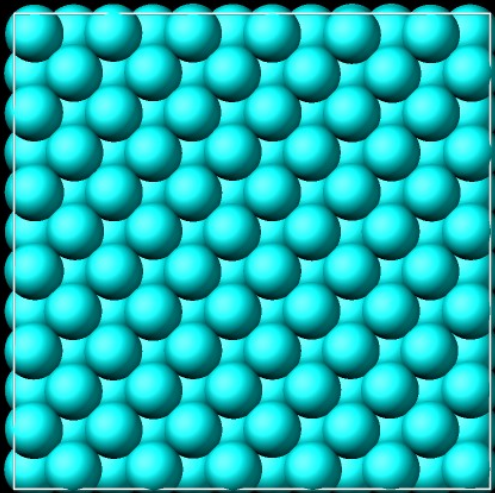
**set**

group all type/fraction 2 0.05 46415



Seed

```
set group all type/fraction 2 0.05 46415
```



# Induce a shearing

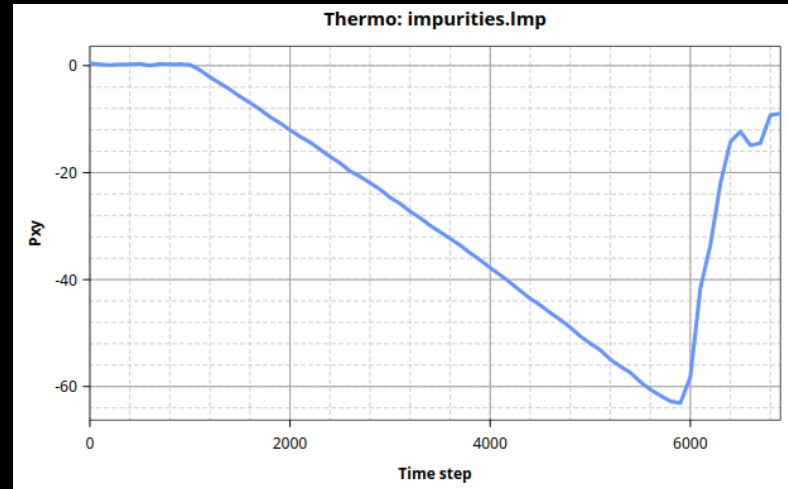
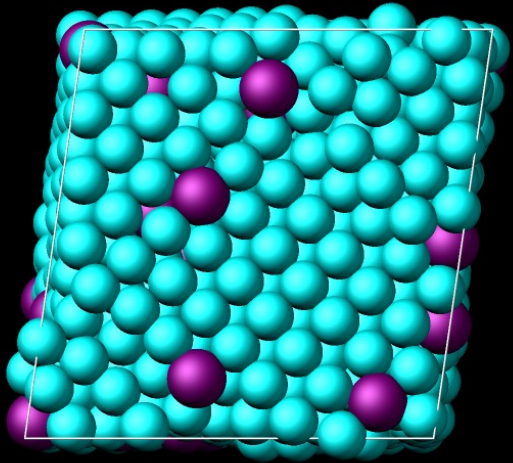
```
run          1000 # System equilibration  
fix          shear    all          deform 1 xy erate 0.01  
run          20000 # Shear run
```

# Induce a shearing

```
run          1000 # System equilibration      Shear rate
fix          shear    all          deform 1 xy erate 0.01
run          20000 # Shear run
```

# Induce a shearing

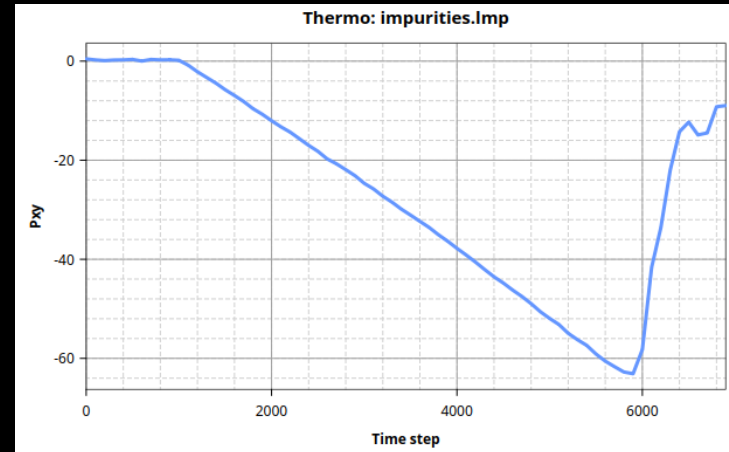
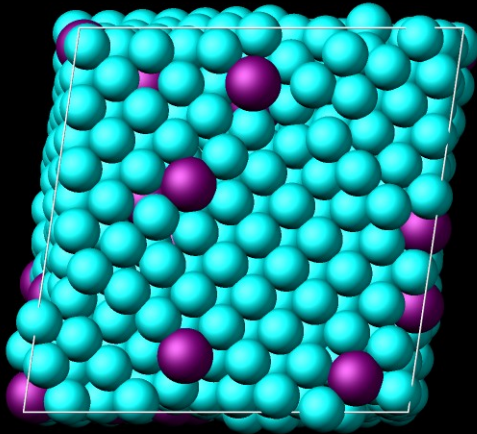
```
run          1000 # System equilibration  
fix          shear    all      deform 1 xy erate 0.01  
run          20000 # Shear run
```

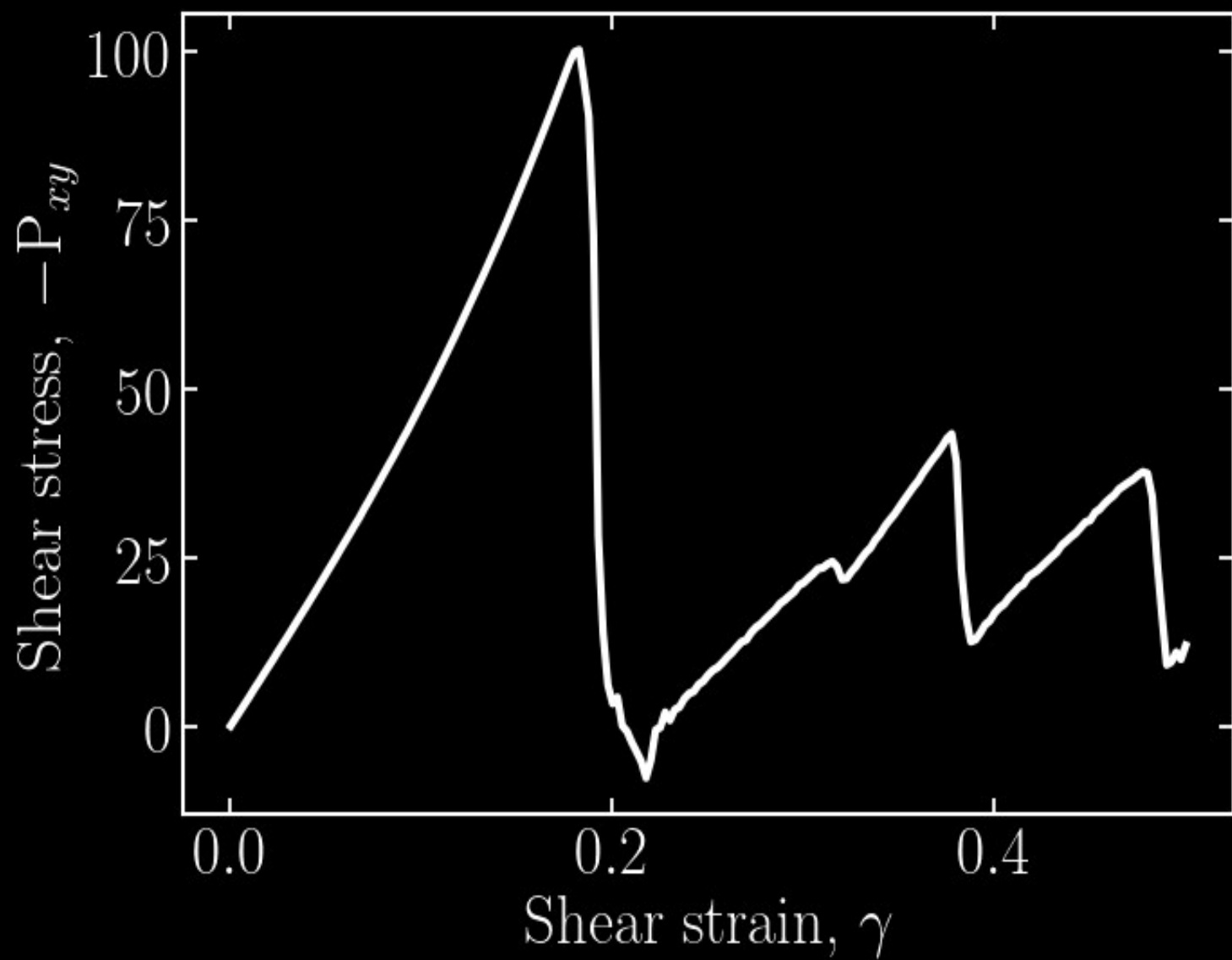


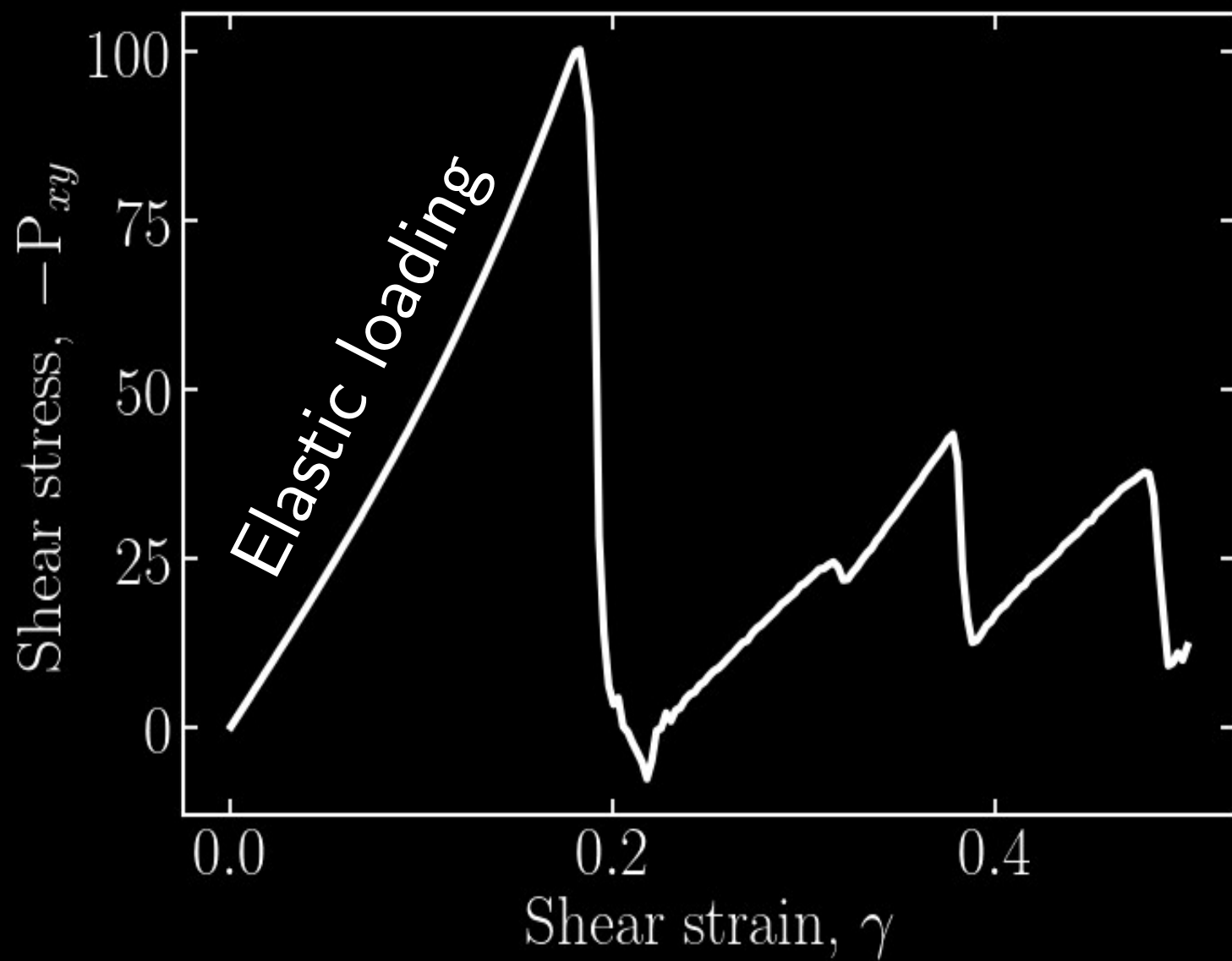
# Induce a shearing

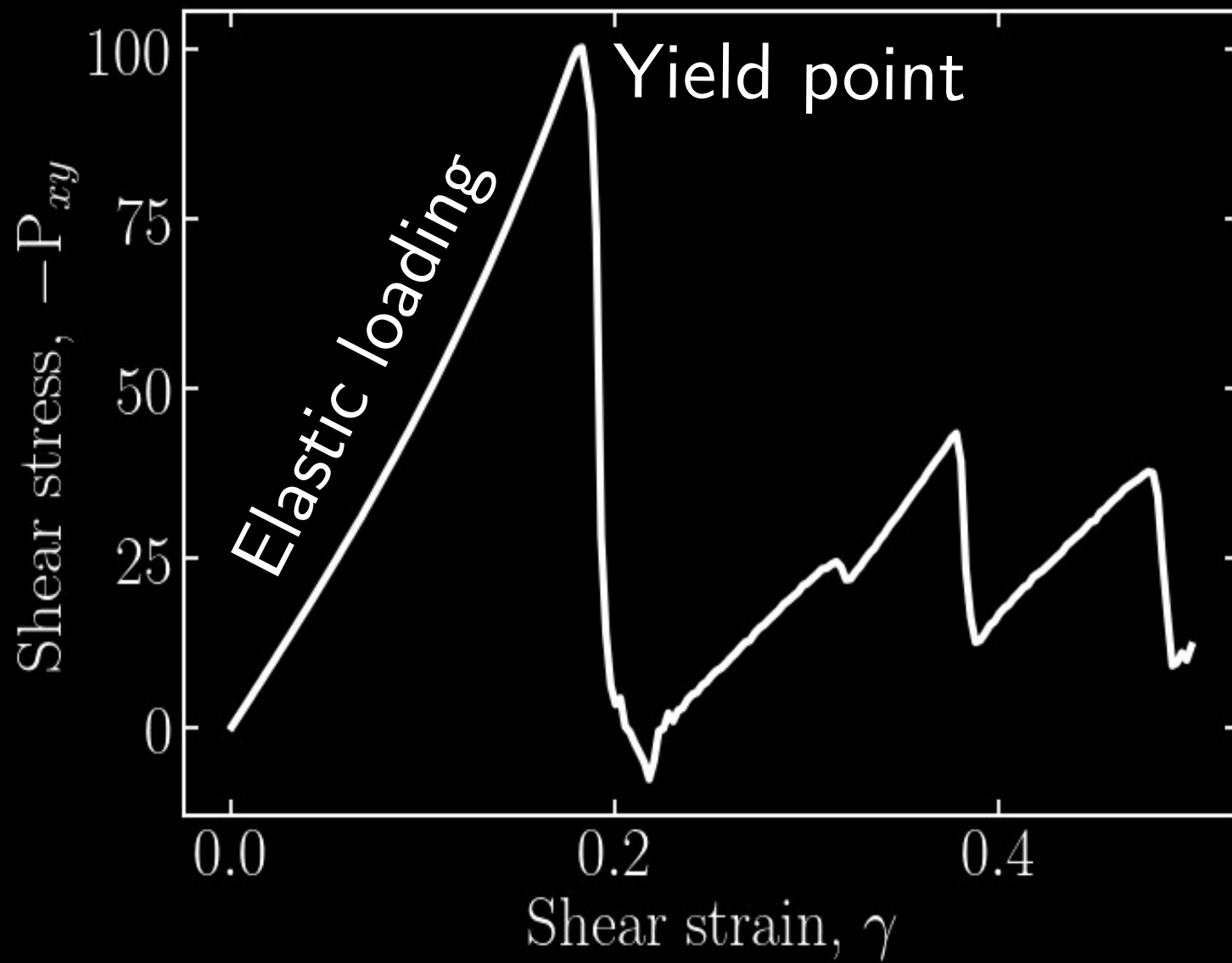
```
run          1000 # System equilibration  
fix          shear    all      deform 1 xy erate 0.01  
variable     stress equal -pxy  
variable     strain equal xy/ly  
fix          myav1 all ave/time 100 1 100 v_strain v_stress file stress_strain_0.05.dat  
run          20000 # Shear run
```

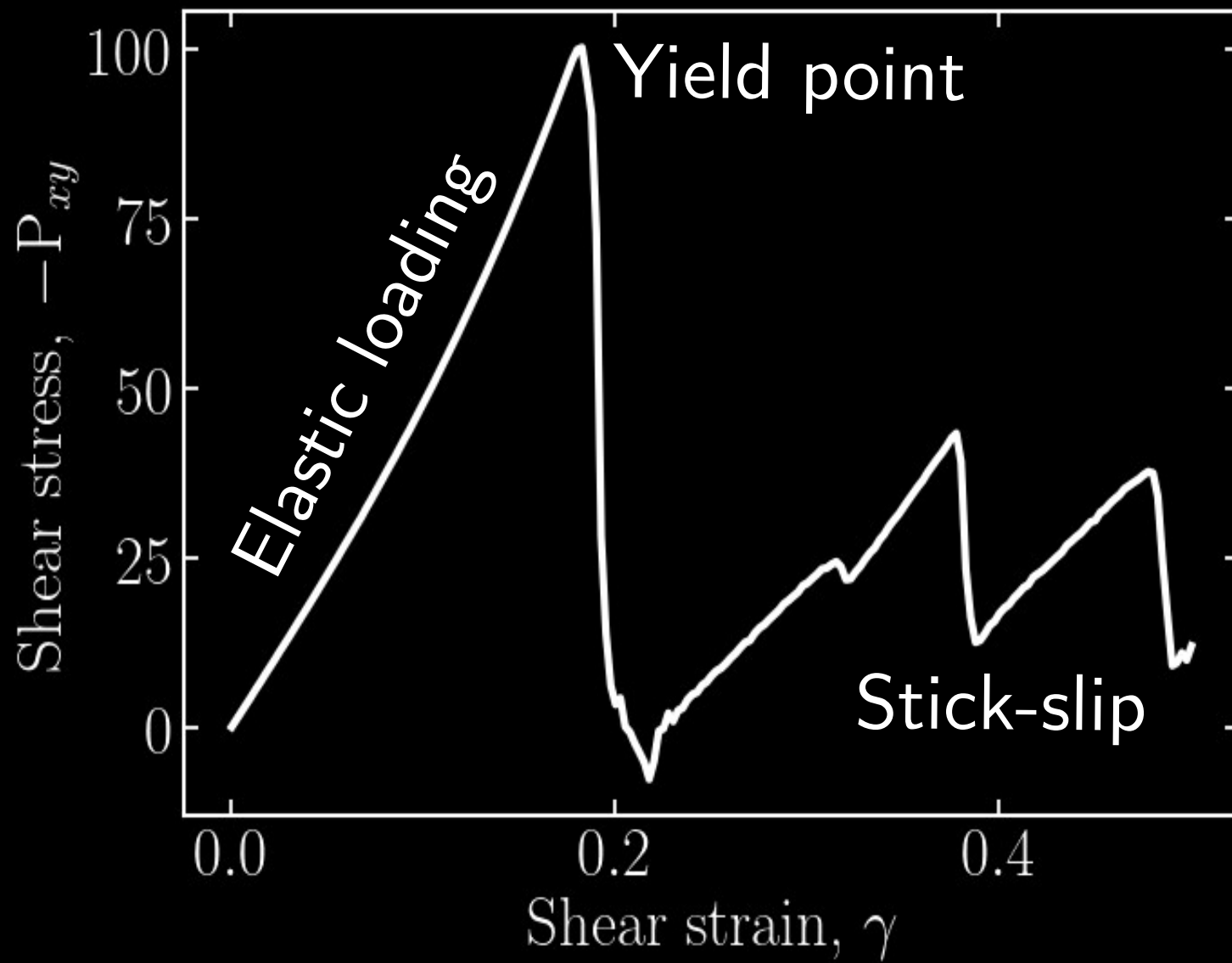
Write to file



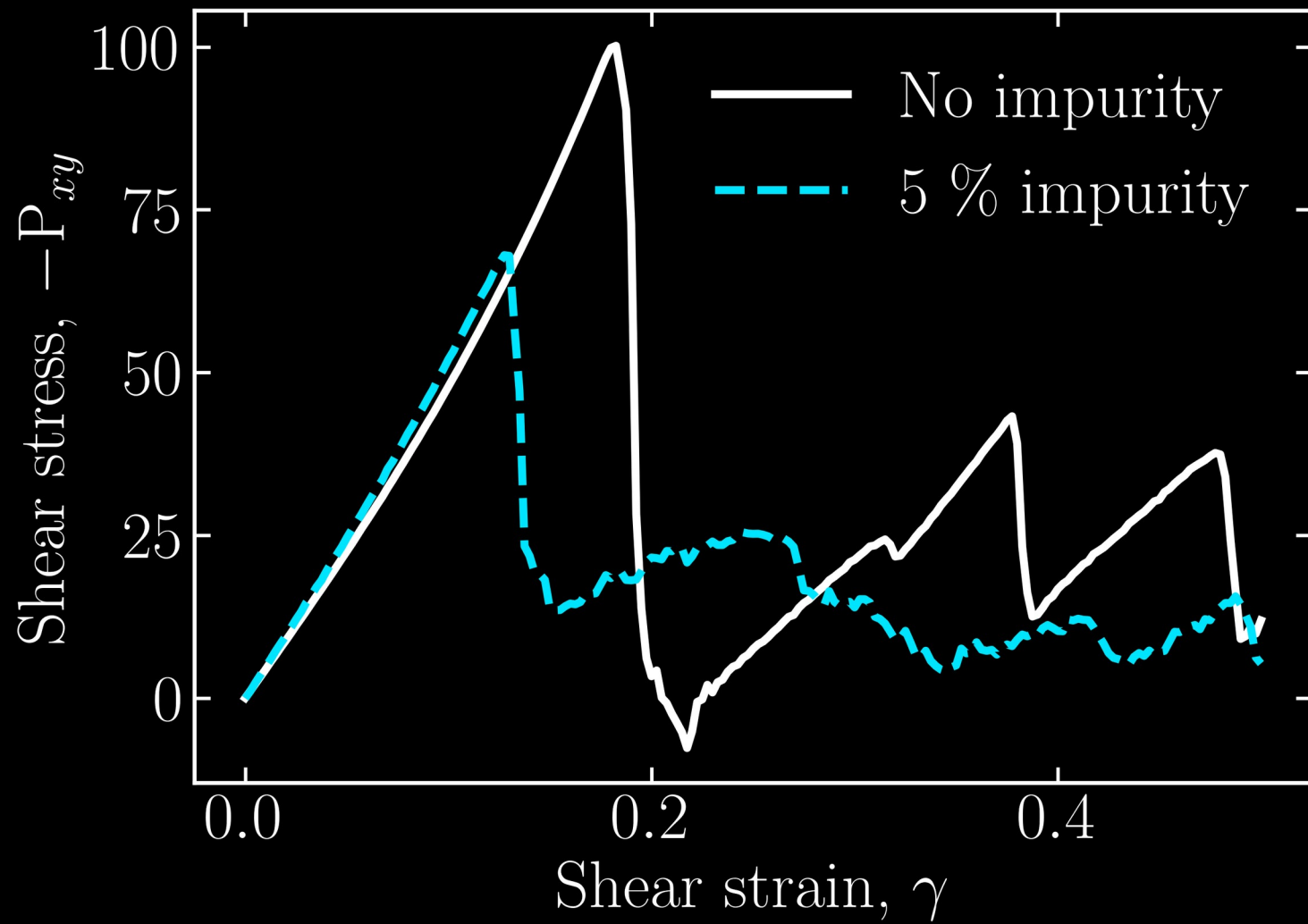






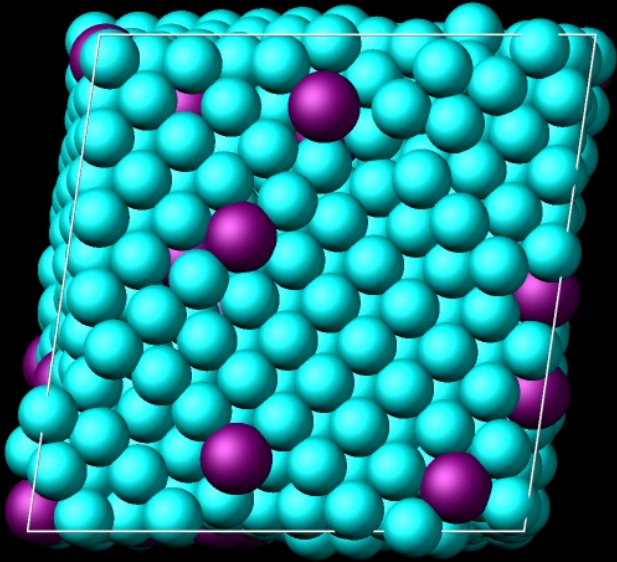


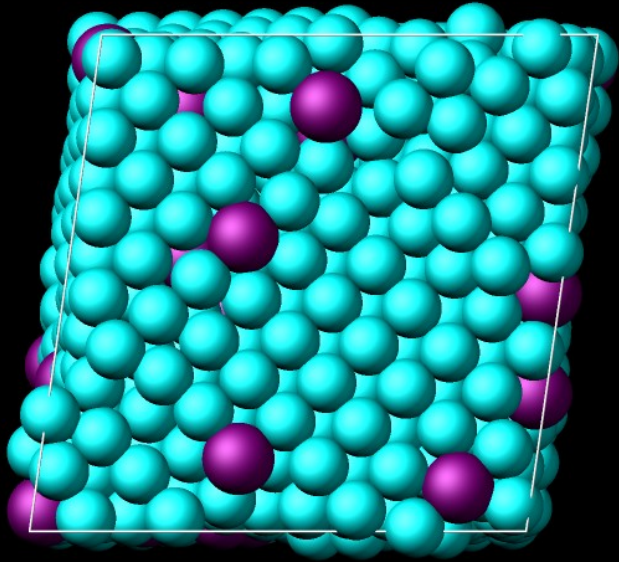
What is the effect of impurity on the stress-strain response ?



Impurity create local strain

→ some sites become unstable earlier





Impurity create local strain

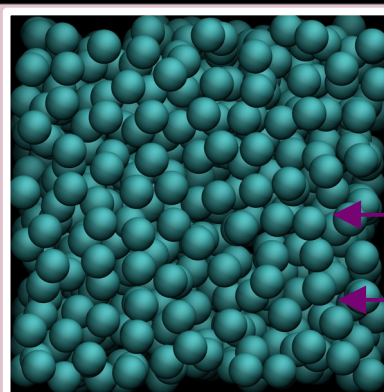
→ some sites become unstable earlier

Crystal planes are less uniform

→ critical stress for slip varies from place to place

→ dislocations nucleate at many sites at different strains

Possibility to use a barostat



$$T_0 = 0.8$$

$$P_0 = 0.8$$

Volume

Energy

$$\mathbf{V}_i \rightarrow \mu \mathbf{V}_i$$

$$\mu = 1 - \frac{\Delta t}{\tau} \kappa_T (P_0 - P)$$

```

16 fix          mynve      all      nve # Integrate the equations of motion
17 timestep      0.0025    # Set the integration timestep to 0.0025
18 fix          myber1     all      temp/berendsen 0.8 0.8 0.1 # Set the temperature
19 fix          myber2     all      press/berendsen iso 1.0 1.0 1 # Set the pressure

```

